

Radiation Pressure Model

5.4

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Module Index

1.1 Modules

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Chapter 2

Namespace Index

2.1 Namespace List

Here is a list of all namespaces with brief descriptions:

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Provide information on bodies that may cause shadowing or reflected illumination	76

Chapter 5

File Index

5.1 File List

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Chapter 6

Module Documentation

6.1 Models

Modules

- [Interactions](#)

6.1.1 Detailed Description

6.2 Interactions

Modules

- [RadiationPressure](#)

6.2.1 Detailed Description

6.3 RadiationPressure

Files

- file [flat_plate_radiation_facet.hh](#)
Individual facets for use with rad environment interaction models.
- file [flat_plate_radiation_factory.hh](#)
Factory that creates an interaction facet, for a specific environment interaction model, from a facet model.
- file [radiation_base_facet.hh](#)
Individual facets for use with radiation environment interaction models.
- file [radiation_default_surface.hh](#)
Individual facets for use with radiation environment interaction models.
- file [radiation_facet.hh](#)
Individual facets for use with radiation environment interaction models.
- file [radiation_messages.hh](#)

- Define the class *RadiationMessages*, the class that specifies the message IDs used in the Radiation model.
- file [radiation_params.hh](#)

A virtual base class for radiation facet parameters, used to create interaction facets for radiation in the *Interaction↔SurfaceFactory*s.
- file [radiation_pressure.hh](#)

Radiation pressure top-level definition.
- file [radiation_source.hh](#)

Radiation pressure parameter and variable definitions.
- file [radiation_surface.hh](#)

Vehicle surface model for general environment interaction models.
- file [radiation_surface_factory.hh](#)

Factory that creates an interaction surface, for a specific environment interaction model, from a surface model.
- file [radiation_third_body.hh](#)

Define the class *RadiationThirdBody*.
- file [flat_plate_radiation_facet.cc](#)

Define member functions for class *FlatPlateRadiationFacet*.
- file [flat_plate_radiation_factory.cc](#)

Factory that creates a *FlatPlateRadiationFacet*, from a facet model.
- file [radiation_base_facet.cc](#)

Define member functions for class *RadiationBaseFacet*.
- file [radiation_default_surface.cc](#)

Default surface for use with Radiation Pressure interaction model.
- file [radiation_facet.cc](#)

Individual facets for use with Radiation Pressure interaction models.
- file [radiation_messages.cc](#)

Implement the class *RadiationMessages*.
- file [radiation_pressure.cc](#)

Calculation of force and torque due to radiation pressure.
- file [radiation_pressure__default_surface.cc](#)

Calculation of force and torque due to radiation pressure.
- file [radiation_pressure__surface_model.cc](#)

Calculation of force and torque due to radiation pressure.
- file [radiation_source.cc](#)

Definition of methods associated with Primary Sources.
- file [radiation_surface.cc](#)

Vehicle surface model for Radiation Pressure model.
- file [radiation_surface_factory.cc](#)

Factory that creates an interaction surface, for a specific environment interaction model, from a surface model.
- file [radiation_third_body.cc](#)

Provides the functionality associated with *RadiationThirdBodies*.

6.3.1 Detailed Description

Chapter 7

Namespace Documentation

7.1 jeod Namespace Reference

Namespace jeod.

Data Structures

- class [FlatPlateRadiationFacet](#)
A flat plate facet to be used for radiation interaction.
- class [FlatPlateRadiationFactory](#)
The factory for building flat plate radiation facets.
- class [RadiationBaseFacet](#)
Generic type of facet for radiation pressure.
- class [RadiationDefaultSurface](#)
The default spherical surface for radiation pressure.
- class [RadiationFacet](#)
Generic type of facet for radiation pressure.
- class [RadiationMessages](#)
Provides error messages.
- class [RadiationParams](#)
Provides a parameter list to each facet, based on the facet material properties.
- class [RadiationPressure](#)
Radiation pressure top-level definition.
- class [RadiationSource](#)
Provides information on the source of the incident radiation.
- class [RadiationSurface](#)
The surface of the vehicle that interacts with the incident flux.
- class [RadiationSurfaceFactory](#)
The factory for creating Radiation Surfaces.
- class [RadiationThirdBody](#)
Provide information on bodies that may cause shadowing or reflected illumination.

7.1.1 Detailed Description

Namespace jeod.

Chapter 8

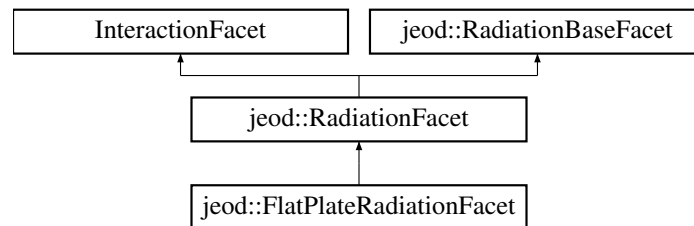
Data Structure Documentation

8.1 jeod::FlatPlateRadiationFacet Class Reference

A flat plate facet to be used for radiation interaction.

```
#include <flat_plate_radiation_facet.hh>
```

Inheritance diagram for jeod::FlatPlateRadiationFacet:



Public Member Functions

- [FlatPlateRadiationFacet](#) ()=default
- [~FlatPlateRadiationFacet](#) () override=default
- [FlatPlateRadiationFacet & operator=](#) (const [FlatPlateRadiationFacet](#) &)=delete
- [FlatPlateRadiationFacet](#) (const [FlatPlateRadiationFacet](#) &)=delete
- void [incident_radiation](#) (const double flux_mag, const double flux_struct_hat[3], const bool calculate_forces) override
 - Calculation of force and torque due to radiation pressure.*
- void [initialize_geom](#) (double center_grav[3]) override
 - Initializes the Facet for use in the model.*
- void [define_facet](#) (FlatPlate *flat_plate)
 - Defines the facet data values.*
- void [radiation_pressure](#) () override
 - Calculates the radiative emission force, accumulated force, and torque acting on a facet.*

Data Fields

- double * [normal](#) {}
Unit vector normal to the plate surface, pointing outward (structural frame).
- double [incident_flux_hat](#) [3] {}
Temporary value.

Private Member Functions

- [JEOD_CLASS_ESTABLISH_FRIENDS2](#) (jeod, [FlatPlateRadiationFacet](#))

Private Attributes

- double [sin_theta](#) {}
Theta is the angle between the plate and the radiation vector.

Additional Inherited Members

8.1.1 Detailed Description

A flat plate facet to be used for radiation interaction.

Definition at line 86 of file `flat_plate_radiation_facet.hh`.

8.1.2 Constructor & Destructor Documentation

8.1.2.1 FlatPlateRadiationFacet() [1/2]

```
jeod::FlatPlateRadiationFacet::FlatPlateRadiationFacet ( ) [default]
```

8.1.2.2 ~FlatPlateRadiationFacet()

```
jeod::FlatPlateRadiationFacet::~~FlatPlateRadiationFacet ( ) [override], [default]
```

8.1.2.3 FlatPlateRadiationFacet() [2/2]

```
jeod::FlatPlateRadiationFacet::FlatPlateRadiationFacet (
    const FlatPlateRadiationFacet & ) [delete]
```

8.1.3 Member Function Documentation

8.1.3.1 define_facet()

```
void jeod::FlatPlateRadiationFacet::define_facet (
    FlatPlate * flat_plate )
```

Defines the facet data values.

Parameters

in	<i>flat_plate</i>	pointer to the flat plate object
----	-------------------	----------------------------------

Definition at line 54 of file flat_plate_radiation_facet.cc.

References [jeod::RadiationFacet::center_pressure](#), and [normal](#).

Referenced by [jeod::FlatPlateRadiationFactory::create_facet\(\)](#).

8.1.3.2 incident_radiation()

```
void jeod::FlatPlateRadiationFacet::incident_radiation (
    const double flux_mag,
    const double flux_struct_hat[3],
    const bool calculate_forces ) [override], [virtual]
```

Calculation of force and torque due to radiation pressure.

Assumptions and Limitations

- Only called when `flux_mag > 0`

Parameters

in	<i>flux_mag</i>	incident flux (per unit area) Units: N/m
in	<i>flux_struct_hat</i>	the flux unit vector in structural frame
in	<i>calculate_forces</i>	on/off flag for whether to calculate forces.

Implements [jeod::RadiationBaseFacet](#).

Definition at line 85 of file flat_plate_radiation_facet.cc.

References [jeod::RadiationBaseFacet::albedo](#), [jeod::RadiationBaseFacet::areaxflux](#), [jeod::RadiationBaseFacet::areaxflux_e](#), [jeod::RadiationBaseFacet::cx_area](#), [jeod::RadiationBaseFacet::diffuse](#), [jeod::RadiationBaseFacet::F_absorption](#), [jeod::RadiationBaseFacet::F_diffuse](#), [jeod::RadiationBaseFacet::F_specular](#), [normal](#), [sin_theta](#), [jeod::RadiationBaseFacet::speed_of_light](#), [jeod::RadiationBaseFacet::thermal](#), and [jeod::RadiationFacet::two thirds](#).

8.1.3.3 initialize_geom()

```
void jeod::FlatPlateRadiationFacet::initialize_geom (
    double center_grav[3] ) [override], [virtual]
```

Initializes the Facet for use in the model.

Parameters

in	<i>center_grav</i>	center of gravity position Units: M
----	--------------------	--

Implements [jeod::RadiationFacet](#).

Definition at line 68 of file flat_plate_radiation_facet.cc.

References [jeod::RadiationFacet::center_pressure](#), [jeod::RadiationFacet::crot_to_cp](#), [jeod::RadiationFacet::initialize\(\)](#), and [jeod::RadiationBaseFacet::thermal](#).

8.1.3.4 JEOD_CLASS_ESTABLISH_FRIENDS2()

```
jeod::FlatPlateRadiationFacet::JEOD_CLASS_ESTABLISH_FRIENDS2 (
    jeod ,
    FlatPlateRadiationFacet ) [private]
```

8.1.3.5 operator=()

```
FlatPlateRadiationFacet& jeod::FlatPlateRadiationFacet::operator= (
    const FlatPlateRadiationFacet & ) [delete]
```

8.1.3.6 radiation_pressure()

```
void jeod::FlatPlateRadiationFacet::radiation_pressure ( ) [override], [virtual]
```

Calculates the radiative emission force, accumulated force, and torque acting on a facet.

Implements [jeod::RadiationFacet](#).

Definition at line 135 of file flat_plate_radiation_facet.cc.

References [jeod::RadiationFacet::crot_to_cp](#), [jeod::RadiationBaseFacet::F_absorption](#), [jeod::RadiationBaseFacet::F_diffuse](#), [jeod::RadiationBaseFacet::F_emission](#), [jeod::RadiationBaseFacet::F_specular](#), [normal](#), [jeod::RadiationBaseFacet::speed_of_light](#), [jeod::RadiationBaseFacet::thermal](#), [jeod::RadiationFacet::two_thirds](#), and [jeod::RadiationMessages::unknown_numerical_error](#).

8.1.4 Field Documentation

8.1.4.1 incident_flux_hat

```
double jeod::FlatPlateRadiationFacet::incident_flux_hat[3] {}
```

Temporary value.

trick_units(-)

Definition at line 101 of file flat_plate_radiation_facet.hh.

8.1.4.2 normal

```
double* jeod::FlatPlateRadiationFacet::normal {}
```

Unit vector normal to the plate surface, pointing outward (structural frame).

once the radiation surface is initialized, it points to the normal found in FlatPlate trick_units(-)

Definition at line 96 of file flat_plate_radiation_facet.hh.

Referenced by define_facet(), incident_radiation(), and radiation_pressure().

8.1.4.3 sin_theta

```
double jeod::FlatPlateRadiationFacet::sin_theta {} [private]
```

Theta is the angle between the plate and the radiation vector.

trick_units(-)

Definition at line 107 of file flat_plate_radiation_facet.hh.

Referenced by incident_radiation().

The documentation for this class was generated from the following files:

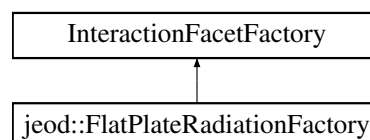
- [flat_plate_radiation_facet.hh](#)
- [flat_plate_radiation_facet.cc](#)

8.2 jeod::FlatPlateRadiationFactory Class Reference

The factory for building flat plate radiation facets.

```
#include <flat_plate_radiation_factory.hh>
```

Inheritance diagram for jeod::FlatPlateRadiationFactory:



Public Member Functions

- [FlatPlateRadiationFactory](#) ()
Constructor for [FlatPlateRadiationFactory](#).
- [~FlatPlateRadiationFactory](#) () override=default
- [FlatPlateRadiationFactory](#) & [operator=](#) (const [FlatPlateRadiationFactory](#) &)=delete
- [FlatPlateRadiationFactory](#) (const [FlatPlateRadiationFactory](#) &)=delete
- InteractionFacet * [create_facet](#) (Facet *facet, FacetParams *params) override
Records the data for the Flat Plate Radiation Facet.
- bool [is_correct_factory](#) (Facet *facet) override
Tests to ensure that the factory can function on the facet as intended.

Private Member Functions

- JEOD_CLASS_ESTABLISH_FRIENDS2 p (jeod, [FlatPlateRadiationFactory](#))

8.2.1 Detailed Description

The factory for building flat plate radiation facets.

Definition at line 84 of file flat_plate_radiation_factory.hh.

8.2.2 Constructor & Destructor Documentation

8.2.2.1 FlatPlateRadiationFactory() [1/2]

```
jeod::FlatPlateRadiationFactory::FlatPlateRadiationFactory ( )
```

Constructor for [FlatPlateRadiationFactory](#).

Definition at line 59 of file flat_plate_radiation_factory.cc.

8.2.2.2 ~FlatPlateRadiationFactory()

```
jeod::FlatPlateRadiationFactory::~~FlatPlateRadiationFactory ( ) [override], [default]
```

8.2.2.3 FlatPlateRadiationFactory() [2/2]

```
jeod::FlatPlateRadiationFactory::FlatPlateRadiationFactory (
    const FlatPlateRadiationFactory & ) [delete]
```


8.2.3 Member Function Documentation

8.2.3.1 create_facet()

```
InteractionFacet * jeod::FlatPlateRadiationFactory::create_facet (
    Facet * facet,
    FacetParams * params ) [override]
```

Records the data for the Flat Plate Radiation Facet.

Returns

pointer to the interaction facet that this function creates.

Parameters

in	<i>facet</i>	pointer to the facet
in	<i>params</i>	pointer to the set of parameters for the facet.

Definition at line 70 of file flat_plate_radiation_factory.cc.

References `jeod::FlatPlateRadiationFacet::define_facet()`, and `jeod::RadiationMessages::invalid_setup_error`.

8.2.3.2 is_correct_factory()

```
bool jeod::FlatPlateRadiationFactory::is_correct_factory (
    Facet * facet ) [override]
```

Tests to ensure that the factory can function on the facet as intended.

Returns

Boolean, is this the correct factory?

Parameters

in	<i>facet</i>	pointer to the facet being manipulated by the factory
----	--------------	---

Definition at line 130 of file flat_plate_radiation_factory.cc.

8.2.3.3 operator=()

```
FlatPlateRadiationFactory& jeod::FlatPlateRadiationFactory::operator= (
```

```
const FlatPlateRadiationFactory & ) [delete]
```

8.2.3.4 p()

```
JEOD_CLASS_ESTABLISH_FRIENDS2 jeod::FlatPlateRadiationFactory::p (
    jeod ,
    FlatPlateRadiationFactory ) [private]
```

The documentation for this class was generated from the following files:

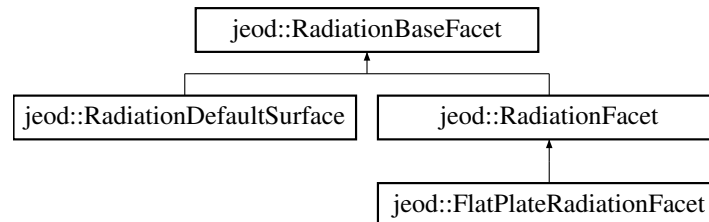
- [flat_plate_radiation_factory.hh](#)
- [flat_plate_radiation_factory.cc](#)

8.3 jeod::RadiationBaseFacet Class Reference

Generic type of facet for radiation pressure.

```
#include <radiation_base_facet.hh>
```

Inheritance diagram for jeod::RadiationBaseFacet:



Public Member Functions

- [RadiationBaseFacet](#) ()=default
- virtual [~RadiationBaseFacet](#) ()=default
- [RadiationBaseFacet & operator=](#) (const [RadiationBaseFacet](#) &)=delete
- [RadiationBaseFacet](#) (const [RadiationBaseFacet](#) &)=delete
- virtual void [initialize](#) ()
initializes the base surface
- virtual void [interact_with_third_body](#) ([RadiationThirdBody](#) *third_body, const bool calc_forces)
Calculation of force, torques and irradiance due to ThirdBody flux.
- virtual void [initialize_runtime_values](#) ()
To initialize the values during each update run.
- virtual void [incident_radiation](#) (const double flux_mag, const double flux_hat[3], const bool calc_forc)=0
Calculates the effect on the facet of the incident radiation.

Data Fields

- double [albedo](#) {-1.0}
Usable value of albedo, set to either `albedo_IR` or `albedo_vis`, depending on situation.
- double [albedo_vis](#) {-1.0}
Fraction of incident visible radiation that is immediately reflected.
- double [albedo_IR](#) {-1.0}
Fraction of incident IR radiation that is immediately reflected.
- double [diffuse](#) {-1.0}
Fraction of reflected radiation that is reflected diffusely (balance reflected specularly)
- ThermalFacetRider [thermal](#)
thermal characteristics of the facet.
- double [cx_area](#) {}
cross-sectional area projected perpendicular to the radiation vector.
- double [areaxflux](#) {}
product of momentum flux and cross-sectional area (`cx_area`).
- double [areaxflux_e](#) {}
product of energy flux and cross-sectional area
- double [F_absorption](#) [3] {}
Force due to photon absorption from ONLY ONE source.
- double [F_specular](#) [3] {}
Force due to photon specular reflection from ONLY ONE source.
- double [F_diffuse](#) [3] {}
Force due to photon diffuse reflection from ONLY ONE source.
- double [F_emission](#) [3] {}
Force due to photon (thermal) emission.

Static Protected Attributes

- static constexpr double [speed_of_light](#) {299792458.0}
Speed of light in vacuum.

Private Member Functions

- JEOD_CLASS_ESTABLISH_FRIENDS2 [p](#) (jeod, [RadiationBaseFacet](#))

8.3.1 Detailed Description

Generic type of facet for radiation pressure.

Definition at line 86 of file `radiation_base_facet.hh`.

8.3.2 Constructor & Destructor Documentation

8.3.2.1 RadiationBaseFacet() [1/2]

```
jeod::RadiationBaseFacet::RadiationBaseFacet ( ) [default]
```

8.3.2.2 ~RadiationBaseFacet()

```
virtual jeod::RadiationBaseFacet::~~RadiationBaseFacet ( ) [virtual], [default]
```

8.3.2.3 RadiationBaseFacet() [2/2]

```
jeod::RadiationBaseFacet::RadiationBaseFacet (
    const RadiationBaseFacet & ) [delete]
```

8.3.3 Member Function Documentation

8.3.3.1 incident_radiation()

```
virtual void jeod::RadiationBaseFacet::incident_radiation (
    const double flux_mag,
    const double flux_hat[3],
    const bool calc_forc ) [pure virtual]
```

Calculates the effect on the facet of the incident radiation.

Parameters

in	<i>flux_mag</i>	Magnitude of the incident flux
in	<i>flux_hat</i>	unit vector associated with the incident flux vector
in	<i>calc_forc</i>	true/false, do forces get calculated

Implemented in [jeod::FlatPlateRadiationFacet](#), and [jeod::RadiationDefaultSurface](#).

Referenced by [jeod::RadiationSurface::incident_radiation\(\)](#).

8.3.3.2 initialize()

```
void jeod::RadiationBaseFacet::initialize ( ) [virtual]
```

initializes the base surface

Reimplemented in [jeod::RadiationFacet](#), and [jeod::RadiationDefaultSurface](#).

Definition at line 50 of file radiation_base_facet.cc.

References [albedo](#), [albedo_IR](#), [albedo_vis](#), and [jeod::RadiationMessages::invalid_setup_error](#).

Referenced by [jeod::RadiationDefaultSurface::initialize\(\)](#), and [jeod::RadiationFacet::initialize\(\)](#).

8.3.3.3 initialize_runtime_values()

```
void jeod::RadiationBaseFacet::initialize_runtime_values ( ) [virtual]
```

To initialize the values during each update run.

Definition at line 110 of file radiation_base_facet.cc.

References [F_absorption](#), [F_diffuse](#), [F_specular](#), and [thermal](#).

Referenced by [jeod::RadiationSurface::initialize_runtime_values\(\)](#), and [jeod::RadiationPressure::update_default↵_surface\(\)](#).

8.3.3.4 interact_with_third_body()

```
void jeod::RadiationBaseFacet::interact_with_third_body (
    RadiationThirdBody * third_body_ptr,
    const bool calculate_forces ) [virtual]
```

Calculation of force, torques and irradiance due to ThirdBody flux.

Parameters

<i>in, out</i>	<i>third_body_ptr</i>	Third body that emits radiation
<i>in</i>	<i>calculate_forces</i>	Calculate forces on this facet if true.

Definition at line 95 of file radiation_base_facet.cc.

References [jeod::RadiationThirdBody::accumulate_rad_flux\(\)](#), [jeod::RadiationThirdBody::accumulate_refl_flux\(\)](#), [albedo](#), [albedo_IR](#), and [albedo_vis](#).

Referenced by [jeod::RadiationSurface::interact_with_third_body\(\)](#), and [jeod::RadiationPressure::update_default↵_surface\(\)](#).

8.3.3.5 operator=()

```
RadiationBaseFacet& jeod::RadiationBaseFacet::operator= (
    const RadiationBaseFacet & ) [delete]
```

8.3.3.6 p()

```
JEOD_CLASS_ESTABLISH_FRIENDS2 jeod::RadiationBaseFacet::p (
    jeod ,
    RadiationBaseFacet ) [private]
```

8.3.4 Field Documentation

8.3.4.1 albedo

```
double jeod::RadiationBaseFacet::albedo {-1.0}
```

Usable value of albedo, set to either albedo_IR or albedo_vis, depending on situation.

trick_units(-)

Definition at line 95 of file radiation_base_facet.hh.

Referenced by jeod::RadiationFacet::define_facet_core(), jeod::RadiationDefaultSurface::incident_radiation(), jeod::FlatPlateRadiationFacet::incident_radiation(), initialize(), jeod::RadiationDefaultSurface::initialize(), jeod::RadiationFacet::initialize(), and interact_with_third_body().

8.3.4.2 albedo_IR

```
double jeod::RadiationBaseFacet::albedo_IR {-1.0}
```

Fraction of incident IR radiation that is immediately reflected.

trick_units(-)

Definition at line 105 of file radiation_base_facet.hh.

Referenced by initialize(), jeod::RadiationFacet::initialize(), and interact_with_third_body().

8.3.4.3 albedo_vis

```
double jeod::RadiationBaseFacet::albedo_vis {-1.0}
```

Fraction of incident visible radiation that is immediately reflected.

trick_units(-)

Definition at line 100 of file radiation_base_facet.hh.

Referenced by initialize(), jeod::RadiationFacet::initialize(), and interact_with_third_body().

8.3.4.4 areaxflux

```
double jeod::RadiationBaseFacet::areaxflux {}
```

product of momentum flux and cross-sectional area (cx_area).

Highly variable. trick_units(-)

Definition at line 132 of file radiation_base_facet.hh.

Referenced by jeod::RadiationDefaultSurface::incident_radiation(), and jeod::FlatPlateRadiationFacet::incident_↔ radiation().

8.3.4.5 areaxflux_e

```
double jeod::RadiationBaseFacet::areaxflux_e {}
```

product of energy flux and cross-sectional area

trick_units(-)

Definition at line 137 of file radiation_base_facet.hh.

Referenced by jeod::RadiationDefaultSurface::incident_radiation(), and jeod::FlatPlateRadiationFacet::incident_↔ radiation().

8.3.4.6 cx_area

```
double jeod::RadiationBaseFacet::cx_area {}
```

cross-sectional area projected perpendicular to the radiation vector.

For facets that have an orientation, this is intended to be a variable value, dependent upon the orientation. For spherical surfaces, this can be set at initialization. trick_units(m2)

Definition at line 124 of file radiation_base_facet.hh.

Referenced by jeod::RadiationDefaultSurface::incident_radiation(), jeod::FlatPlateRadiationFacet::incident_↔ radiation(), and jeod::RadiationDefaultSurface::initialize().

8.3.4.7 diffuse

```
double jeod::RadiationBaseFacet::diffuse {-1.0}
```

Fraction of reflected radiation that is reflected diffusely (balance reflected specularly)

trick_units(-)

Definition at line 111 of file radiation_base_facet.hh.

Referenced by jeod::RadiationFacet::define_facet_core(), jeod::RadiationDefaultSurface::incident_radiation(), jeod::FlatPlateRadiationFacet::incident_radiation(), jeod::RadiationDefaultSurface::initialize(), and jeod::RadiationFacet::initialize().

8.3.4.8 F_absorption

```
double jeod::RadiationBaseFacet::F_absorption[3] {}
```

Force due to photon absorption from ONLY ONE source.

trick_units(-)

Definition at line 142 of file radiation_base_facet.hh.

Referenced by jeod::RadiationDefaultSurface::incident_radiation(), jeod::FlatPlateRadiationFacet::incident_radiation(), initialize_runtime_values(), jeod::FlatPlateRadiationFacet::radiation_pressure(), and jeod::RadiationPressure::update_default_surface().

8.3.4.9 F_diffuse

```
double jeod::RadiationBaseFacet::F_diffuse[3] {}
```

Force due to photon diffuse reflection from ONLY ONE source.

trick_units(-)

Definition at line 152 of file radiation_base_facet.hh.

Referenced by jeod::RadiationDefaultSurface::incident_radiation(), jeod::FlatPlateRadiationFacet::incident_radiation(), initialize_runtime_values(), jeod::FlatPlateRadiationFacet::radiation_pressure(), and jeod::RadiationPressure::update_default_surface().

8.3.4.10 F_emission

```
double jeod::RadiationBaseFacet::F_emission[3] {}
```

Force due to photon (thermal) emission.

trick_units(-)

Definition at line 157 of file radiation_base_facet.hh.

Referenced by jeod::FlatPlateRadiationFacet::radiation_pressure().

8.3.4.11 F_specular

```
double jeod::RadiationBaseFacet::F_specular[3] {}
```

Force due to photon specular reflection from ONLY ONE source.

trick_units(-)

Definition at line 147 of file radiation_base_facet.hh.

Referenced by jeod::RadiationDefaultSurface::incident_radiation(), jeod::FlatPlateRadiationFacet::incident_↔ radiation(), initialize_runtime_values(), jeod::FlatPlateRadiationFacet::radiation_pressure(), and jeod::Radiation↔ Pressure::update_default_surface().

8.3.4.12 speed_of_light

```
constexpr double jeod::RadiationBaseFacet::speed_of_light {299792458.0} [static], [constexpr],  
[protected]
```

Speed of light in vacuum.

trick_io(*o) trick_units(m/s)

Definition at line 164 of file radiation_base_facet.hh.

Referenced by jeod::RadiationDefaultSurface::incident_radiation(), jeod::FlatPlateRadiationFacet::incident_↔ radiation(), and jeod::FlatPlateRadiationFacet::radiation_pressure().

8.3.4.13 thermal

ThermalFacetRider jeod::RadiationBaseFacet::thermal

thermal characteristics of the facet.

trick_units(-)

Definition at line 116 of file radiation_base_facet.hh.

Referenced by jeod::RadiationSurface::accumulate_thermal_sources(), jeod::RadiationDefaultSurface::add_thermal_integrator_to(), jeod::RadiationSurface::add_thermal_integrators_to(), jeod::RadiationFacet::define_facet_core(), jeod::RadiationSurface::equalize_absorption_emission(), jeod::RadiationDefaultSurface::incident_radiation(), jeod::FlatPlateRadiationFacet::incident_radiation(), jeod::RadiationDefaultSurface::initialize(), jeod::RadiationFacet::initialize(), jeod::RadiationSurface::initialize(), jeod::FlatPlateRadiationFacet::initialize_geom(), initialize_runtime_values(), jeod::FlatPlateRadiationFacet::radiation_pressure(), jeod::RadiationSurface::thermal_integrator(), and jeod::RadiationDefaultSurface::thermal_update().

The documentation for this class was generated from the following files:

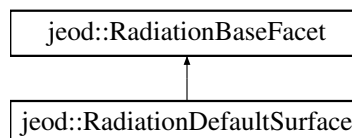
- [radiation_base_facet.hh](#)
- [radiation_base_facet.cc](#)

8.4 jeod::RadiationDefaultSurface Class Reference

The default spherical surface for radiation pressure.

```
#include <radiation_default_surface.hh>
```

Inheritance diagram for jeod::RadiationDefaultSurface:



Public Member Functions

- [RadiationDefaultSurface](#) ()=default
- [~RadiationDefaultSurface](#) () override=default
- [RadiationDefaultSurface](#) & operator= (const [RadiationDefaultSurface](#) &)=delete
- [RadiationDefaultSurface](#) (const [RadiationDefaultSurface](#) &)=delete
- void [initialize](#) () override
 - initializes the default surface*
- void [incident_radiation](#) (const double flux_mag, const double flux_struc_hat[3], const bool calculate_forces) override
 - Calculation of force and torque due to radiation pressure.*
- void [thermal_update](#) ()
 - Test for necessity of maintaining thermal updates, and performs them as necessary.*
- void [add_thermal_integrator_to](#) (DynBody *dyn_body)
 - Adds the thermal integrator of this surface to the integration group of the DynBody.*
- void [set_name](#) (std::string name_in)
 - Setter for the name.*

Data Fields

- double `rad_coeff` {-1.0}
The radiation-equivalent of a drag coefficient.
- double `temperature` {}
The value of the surface kinetic temperature.
- std::string `name`
The name of the surface.
- double `surface_area` {}
surface area of the default sphere.

Private Member Functions

- JEOD_CLASS_ESTABLISH_FRIENDS2 `p` (jeod, `RadiationDefaultSurface`)

Additional Inherited Members

8.4.1 Detailed Description

The default spherical surface for radiation pressure.

Definition at line 89 of file `radiation_default_surface.hh`.

8.4.2 Constructor & Destructor Documentation

8.4.2.1 RadiationDefaultSurface() [1/2]

```
jeod::RadiationDefaultSurface::RadiationDefaultSurface ( ) [default]
```

8.4.2.2 ~RadiationDefaultSurface()

```
jeod::RadiationDefaultSurface::~~RadiationDefaultSurface ( ) [override], [default]
```

8.4.2.3 RadiationDefaultSurface() [2/2]

```
jeod::RadiationDefaultSurface::RadiationDefaultSurface (
    const RadiationDefaultSurface & ) [delete]
```

8.4.3 Member Function Documentation

8.4.3.1 add_thermal_integrator_to()

```
void jeod::RadiationDefaultSurface::add_thermal_integrator_to (
    DynBody * dyn_body )
```

Adds the thermal integrator of this surface to the integration group of the DynBody.

Parameters

<i>in, out</i>	<i>dyn_body</i>	Body associated with this thermal object.
----------------	-----------------	---

Definition at line 268 of file radiation_default_surface.cc.

References `jeod::RadiationBaseFacet::thermal`.

8.4.3.2 incident_radiation()

```
void jeod::RadiationDefaultSurface::incident_radiation (
    const double flux_mag,
    const double flux_struc_hat[3],
    const bool calculate_forces ) [override], [virtual]
```

Calculation of force and torque due to radiation pressure.

Assumptions and Limitations

- Assumes that flux magnitude is positive.
- Will only be called if flux magnitude is positive.
- Assumes spherical surface.

Parameters

<i>in</i>	<i>flux_mag</i>	the magnitude of the incident flux
<i>in</i>	<i>flux_struc_hat</i>	the unit vector in structural frame for the flux vector.
<i>in</i>	<i>calculate_forces</i>	boolean indicating whether to calculate forces.

Implements [jeod::RadiationBaseFacet](#).

Definition at line 214 of file radiation_default_surface.cc.

References `jeod::RadiationBaseFacet::albedo`, `jeod::RadiationBaseFacet::areaxflux`, `jeod::RadiationBaseFacet::areaxflux_e`, `jeod::RadiationBaseFacet::cx_area`, `jeod::RadiationBaseFacet::diffuse`, `jeod::RadiationBaseFacet::F_absorption`, `jeod::RadiationBaseFacet::F_diffuse`, `jeod::RadiationBaseFacet::F_specular`, `jeod::RadiationBaseFacet::speed_of_light`, and `jeod::RadiationBaseFacet::thermal`.

Referenced by `jeod::RadiationPressure::update_default_surface()`.

8.4.3.3 initialize()

```
void jeod::RadiationDefaultSurface::initialize ( ) [override], [virtual]
```

initializes the default surface

Reimplemented from [jeod::RadiationBaseFacet](#).

Definition at line 58 of file radiation_default_surface.cc.

References [jeod::RadiationBaseFacet::albedo](#), [jeod::RadiationBaseFacet::cx_area](#), [jeod::RadiationBaseFacet::diffuse](#), [jeod::RadiationBaseFacet::initialize\(\)](#), [jeod::RadiationMessages::invalid_setup_error](#), [jeod::RadiationMessages::operational_setup_error](#), [rad_coeff](#), [surface_area](#), [temperature](#), and [jeod::RadiationBaseFacet::thermal](#).

Referenced by [jeod::RadiationPressure::initialize\(\)](#).

8.4.3.4 operator=()

```
RadiationDefaultSurface& jeod::RadiationDefaultSurface::operator= (
    const RadiationDefaultSurface & ) [delete]
```

8.4.3.5 p()

```
JEOD_CLASS_ESTABLISH_FRIENDS2 jeod::RadiationDefaultSurface::p (
    jeod ,
    RadiationDefaultSurface ) [private]
```

8.4.3.6 set_name()

```
void jeod::RadiationDefaultSurface::set_name (
    std::string name_in ) [inline]
```

Setter for the name.

Definition at line 148 of file radiation_default_surface.hh.

8.4.3.7 thermal_update()

```
void jeod::RadiationDefaultSurface::thermal_update ( )
```

Test for necessity of maintaining thermal updates, and performs them as necessary.

Definition at line 251 of file radiation_default_surface.cc.

References [temperature](#), and [jeod::RadiationBaseFacet::thermal](#).

Referenced by [jeod::RadiationPressure::update_default_surface\(\)](#).

8.4.4 Field Documentation

8.4.4.1 name

```
std::string jeod::RadiationDefaultSurface::name
```

The name of the surface.

This is optional. `trick_units(-)`

Definition at line 116 of file `radiation_default_surface.hh`.

8.4.4.2 rad_coeff

```
double jeod::RadiationDefaultSurface::rad_coeff {-1.0}
```

The radiation-equivalent of a drag coefficient.

It is the value by which the area must be multiplied in order to generate the same force if the reflecting surface were considered a perfectly absorbing surface. For a sphere, this has a value between 1.0 (perfectly absorbing, or perfect specular reflection) to 1.4444 (all diffuse reflection). Specify either: `rad_coeff` OR (albedo AND diffuse). NOTE 1 - this value is used at initialization only; changes to its value mid-simulation cannot be effected. NOTE 2 - the values albedo and diffuse are inherited from [RadiationBaseFacet](#). `trick_units(-)`

Definition at line 106 of file `radiation_default_surface.hh`.

Referenced by `initialize()`.

8.4.4.3 surface_area

```
double jeod::RadiationDefaultSurface::surface_area {}
```

surface area of the default sphere.

`trick_units(m2)`

Definition at line 125 of file `radiation_default_surface.hh`.

Referenced by `initialize()`.

8.4.4.4 temperature

```
double jeod::RadiationDefaultSurface::temperature {}
```

The value of the surface kinetic temperature.

trick_units(K)

Definition at line 111 of file radiation_default_surface.hh.

Referenced by initialize(), and thermal_update().

The documentation for this class was generated from the following files:

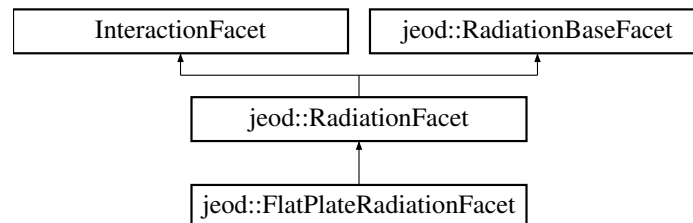
- [radiation_default_surface.hh](#)
- [radiation_default_surface.cc](#)

8.5 jeod::RadiationFacet Class Reference

Generic type of facet for radiation pressure.

```
#include <radiation_facet.hh>
```

Inheritance diagram for jeod::RadiationFacet:



Public Member Functions

- [RadiationFacet](#) ()=default
- [~RadiationFacet](#) () override=default
- [RadiationFacet & operator=](#) (const [RadiationFacet](#) &)=delete
- [RadiationFacet](#) (const [RadiationFacet](#) &)=delete
- virtual void [radiation_pressure](#) ()=0
Calculates the effect of radiation pressure on the facet.
- virtual void [initialize_geom](#) (double cg[3])=0
Initialize the facet geometry.
- void [initialize](#) () override
Run sanity checks on input variables.
- void [define_facet_core](#) (Facet *facet, const ThermalFacetRider &facet_thermal, [RadiationParams](#) *params)
Defines the inherent facet values.
- ThermalIntegrableObject & [get_thermal_integrator](#) ()
Get the integrator for thermal characteristics.

Data Fields

- double `crot_to_cp` [3] {}
position of center of pressure w.r.t.
- double * `center_pressure` {}
Center of pressure (in structural frame).

Static Protected Attributes

- static constexpr double `two_thirds` {2.0 / 3.0}
quite literally, 2/3.

Private Member Functions

- JEOD_CLASS_ESTABLISH_FRIENDS2 `p` (jeod, `RadiationFacet`)

8.5.1 Detailed Description

Generic type of facet for radiation pressure.

Definition at line 87 of file `radiation_facet.hh`.

8.5.2 Constructor & Destructor Documentation

8.5.2.1 `RadiationFacet()` [1/2]

```
jeod::RadiationFacet::RadiationFacet ( ) [default]
```

8.5.2.2 `~RadiationFacet()`

```
jeod::RadiationFacet::~~RadiationFacet ( ) [override], [default]
```

8.5.2.3 `RadiationFacet()` [2/2]

```
jeod::RadiationFacet::RadiationFacet (
    const RadiationFacet & ) [delete]
```

8.5.3 Member Function Documentation

8.5.3.1 `define_facet_core()`

```
void jeod::RadiationFacet::define_facet_core (
    Facet * facet,
    const ThermalFacetRider & facet_thermal,
    RadiationParams * params )
```

Defines the inherent facet values.

Parameters

in	<i>facet</i>	pointer to the facet
in	<i>facet_thermal</i>	reference to the thermal components of the facet.
in	<i>params</i>	pointer to the list of material parameters for the facet.

Definition at line 82 of file radiation_facet.cc.

References [jeod::RadiationBaseFacet::albedo](#), [jeod::RadiationParams::albedo](#), [jeod::RadiationBaseFacet::diffuse](#), [jeod::RadiationParams::diffuse](#), [jeod::RadiationBaseFacet::thermal](#), and [jeod::RadiationParams::thermal](#).

8.5.3.2 get_thermal_integrator()

```
ThermalIntegrableObject& jeod::RadiationFacet::get_thermal_integrator ( ) [inline]
```

Get the integrator for thermal characteristics.

Returns

Integrable object that integrates temperature.

Definition at line 144 of file radiation_facet.hh.

8.5.3.3 initialize()

```
void jeod::RadiationFacet::initialize ( ) [override], [virtual]
```

Run sanity checks on input variables.

Reimplemented from [jeod::RadiationBaseFacet](#).

Definition at line 52 of file radiation_facet.cc.

References [jeod::RadiationBaseFacet::albedo](#), [jeod::RadiationBaseFacet::albedo_IR](#), [jeod::RadiationBaseFacet::albedo_vis](#), [jeod::RadiationBaseFacet::diffuse](#), [jeod::RadiationBaseFacet::initialize\(\)](#), [jeod::RadiationMessages::invalid_setup_error](#), and [jeod::RadiationBaseFacet::thermal](#).

Referenced by [jeod::FlatPlateRadiationFacet::initialize_geom\(\)](#).

8.5.3.4 initialize_geom()

```
virtual void jeod::RadiationFacet::initialize_geom (
    double cg[3] ) [pure virtual]
```

Initialize the facet geometry.

Parameters

<i>cg</i>	Center of mass.
-----------	-----------------

Implemented in [jeod::FlatPlateRadiationFacet](#).

Referenced by [jeod::RadiationSurface::initialize\(\)](#).

8.5.3.5 operator=()

```
RadiationFacet& jeod::RadiationFacet::operator= (
    const RadiationFacet & ) [delete]
```

8.5.3.6 p()

```
JEOD_CLASS_ESTABLISH_FRIENDS2 jeod::RadiationFacet::p (
    jeod ,
    RadiationFacet ) [private]
```

8.5.3.7 radiation_pressure()

```
virtual void jeod::RadiationFacet::radiation_pressure ( ) [pure virtual]
```

Calculates the effect of radiation pressure on the facet.

Implemented in [jeod::FlatPlateRadiationFacet](#).

Referenced by [jeod::RadiationSurface::radiation_pressure\(\)](#).

8.5.4 Field Documentation**8.5.4.1 center_pressure**

```
double* jeod::RadiationFacet::center_pressure { }
```

Center of pressure (in structural frame).

Potentially variable, depending on the topology of the facet. For a flat plate facet, this just points to the position found in [FlatPlate::trick_units\(m\)](#)

Definition at line 106 of file [radiation_facet.hh](#).

Referenced by [jeod::FlatPlateRadiationFacet::define_facet\(\)](#), and [jeod::FlatPlateRadiationFacet::initialize_geom\(\)](#).

8.5.4.2 crot_to_cp

```
double jeod::RadiationFacet::crot_to_cp[3] {}
```

position of center of pressure w.r.t.

center of rotation (usually center of mass, or center of gravity), expressed in structural reference frame. Potentially variable, depending on the topology of the facet and the configuration of the vehicle. `trick_units(m)`

Definition at line 99 of file `radiation_facet.hh`.

Referenced by `jeod::FlatPlateRadiationFacet::initialize_geom()`, and `jeod::FlatPlateRadiationFacet::radiation_↔pressure()`.

8.5.4.3 two_thirds

```
constexpr double jeod::RadiationFacet::two_thirds {2.0 / 3.0} [static], [constexpr], [protected]
```

quite literally, 2/3.

Occurs frequently with diffuse reflection and emission, so it is calculated once. `trick_io(*o) trick_units(–)`

Definition at line 115 of file `radiation_facet.hh`.

Referenced by `jeod::FlatPlateRadiationFacet::incident_radiation()`, and `jeod::FlatPlateRadiationFacet::radiation_↔pressure()`.

The documentation for this class was generated from the following files:

- [radiation_facet.hh](#)
- [radiation_facet.cc](#)

8.6 jeod::RadiationMessages Class Reference

Provides error messages.

```
#include <radiation_messages.hh>
```

Public Member Functions

- [RadiationMessages](#) ()=delete
- [RadiationMessages](#) (const [RadiationMessages](#) &)=delete
- [RadiationMessages](#) & operator= (const [RadiationMessages](#) &)=delete

Static Public Attributes

- static const char * [incomplete_setup_error](#) = "interactions/radiation_pressure/" "incomplete_setup_error"
The setup was not fully defined.
- static const char * [invalid_setup_error](#) = "interactions/radiation_pressure/" "invalid_setup_error"
The setup is invalid.
- static const char * [operational_setup_error](#) = "interactions/radiation_pressure/" "operational_setup_error"
Something internal went wrong in the setup methods without obvious user error.
- static const char * [invalid_function_call](#) = "interactions/radiation_pressure/" "invalid_function_call"
A function was called before it was fully implemented.
- static const char * [unknown_numerical_error](#) = "interactions/radiation_pressure/" "unknown_numerical_error"
Something went horribly wrong.

Private Member Functions

- [JEOD_CLASS_ESTABLISH_FRIENDS2](#) (jeod, [RadiationMessages](#))

8.6.1 Detailed Description

Provides error messages.

Definition at line 84 of file radiation_messages.hh.

8.6.2 Constructor & Destructor Documentation

8.6.2.1 RadiationMessages() [1/2]

```
jeod::RadiationMessages::RadiationMessages ( ) [delete]
```

8.6.2.2 RadiationMessages() [2/2]

```
jeod::RadiationMessages::RadiationMessages (
    const RadiationMessages & ) [delete]
```

8.6.3 Member Function Documentation

8.6.3.1 JEOD_CLASS_ESTABLISH_FRIENDS2()

```
jeod::RadiationMessages::JEOD_CLASS_ESTABLISH_FRIENDS2 (
    jeod ,
    RadiationMessages ) [private]
```

8.6.3.2 operator=()

```
RadiationMessages& jeod::RadiationMessages::operator= (
    const RadiationMessages & ) [delete]
```

8.6.4 Field Documentation

8.6.4.1 incomplete_setup_error

```
char const * jeod::RadiationMessages::incomplete_setup_error = "interactions/radiation_↵
pressure/" "incomplete_setup_error" [static]
```

The setup was not fully defined.

trick_units(-)

Definition at line 92 of file radiation_messages.hh.

Referenced by jeod::RadiationPressure::add_third_body(), and jeod::RadiationSource::initialize().

8.6.4.2 invalid_function_call

```
char const * jeod::RadiationMessages::invalid_function_call = "interactions/radiation_pressure/"
"invalid_function_call" [static]
```

A function was called before it was fully implemented.

trick_units(-)

Definition at line 105 of file radiation_messages.hh.

Referenced by jeod::RadiationPressure::find_third_body(), jeod::RadiationPressure::set_third_body_active(), and jeod::RadiationPressure::set_third_body_inactive().

8.6.4.3 invalid_setup_error

```
char const * jeod::RadiationMessages::invalid_setup_error = "interactions/radiation_pressure/"
"invalid_setup_error" [static]
```

The setup is invalid.

trick_units(-)

Definition at line 96 of file radiation_messages.hh.

Referenced by jeod::RadiationSurfaceFactory::add_facet_params(), jeod::RadiationPressure::add_third_↵
body(), jeod::RadiationSurface::allocate_interaction_facet(), jeod::RadiationThirdBody::calculate_shadow(),
jeod::RadiationThirdBody::convert_shadow_from_int(), jeod::FlatPlateRadiationFactory::create_facet(), jeod::↵
RadiationBaseFacet::initialize(), jeod::RadiationDefaultSurface::initialize(), jeod::RadiationFacet::initialize(), jeod↵
::RadiationSurface::initialize(), jeod::RadiationThirdBody::initialize(), jeod::RadiationThirdBody::process_third_↵
body(), and jeod::RadiationThirdBody::update_third_body_state().

8.6.4.4 operational_setup_error

```
char const * jeod::RadiationMessages::operational_setup_error = "interactions/radiation_↵  
pressure/" "operational_setup_error" [static]
```

Something internal went wrong in the setup methods without obvious user error.

trick_units(-)

Definition at line 101 of file radiation_messages.hh.

Referenced by jeod::RadiationSurface::allocate_array(), jeod::RadiationSurface::allocate_interaction_facet(),
jeod::RadiationDefaultSurface::initialize(), jeod::RadiationSurface::initialize(), and jeod::RadiationThirdBody↵
::process_third_body().

8.6.4.5 unknown_numerical_error

```
char const * jeod::RadiationMessages::unknown_numerical_error = "interactions/radiation_↵  
pressure/" "unknown_numerical_error" [static]
```

Something went horribly wrong.

trick_units(-)

Definition at line 110 of file radiation_messages.hh.

Referenced by jeod::FlatPlateRadiationFacet::radiation_pressure().

The documentation for this class was generated from the following files:

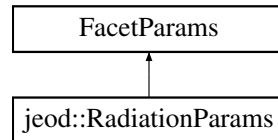
- [radiation_messages.hh](#)
- [radiation_messages.cc](#)

8.7 jeod::RadiationParams Class Reference

Provides a parameter list to each facet, based on the facet material properties.

```
#include <radiation_params.hh>
```

Inheritance diagram for jeod::RadiationParams:



Public Member Functions

- [RadiationParams](#) ()=default
- [~RadiationParams](#) () override=default
- [RadiationParams](#) & [operator=](#) (const [RadiationParams](#) &)=delete
- [RadiationParams](#) (const [RadiationParams](#) &)=delete

Data Fields

- double [albedo](#) {}
Fraction of incident radiation that is immediately reflected.
- double [diffuse](#) {}
Fraction of reflected radiation that is reflected diffusely (balance reflected specularly)
- ThermalParams [thermal](#)
Additional thermal parameters.

Private Member Functions

- JEOD_CLASS_ESTABLISH_FRIENDS2 [p](#) (jeod, [RadiationParams](#))

8.7.1 Detailed Description

Provides a parameter list to each facet, based on the facet material properties.

Definition at line 79 of file radiation_params.hh.

8.7.2 Constructor & Destructor Documentation

8.7.2.1 RadiationParams() [1/2]

```
jeod::RadiationParams::RadiationParams ( ) [default]
```

8.7.2.2 ~RadiationParams()

```
jeod::RadiationParams::~~RadiationParams ( ) [override], [default]
```

8.7.2.3 RadiationParams() [2/2]

```
jeod::RadiationParams::RadiationParams (
    const RadiationParams & ) [delete]
```

8.7.3 Member Function Documentation

8.7.3.1 operator=()

```
RadiationParams& jeod::RadiationParams::operator= (
    const RadiationParams & ) [delete]
```

8.7.3.2 p()

```
JEOD_CLASS_ESTABLISH_FRIENDS2 jeod::RadiationParams::p (
    jeod ,
    RadiationParams ) [private]
```

8.7.4 Field Documentation

8.7.4.1 albedo

```
double jeod::RadiationParams::albedo {}
```

Fraction of incident radiation that is immediately reflected.

trick_units(-)

Definition at line 85 of file radiation_params.hh.

Referenced by jeod::RadiationFacet::define_facet_core().

8.7.4.2 diffuse

```
double jeod::RadiationParams::diffuse {}
```

Fraction of reflected radiation that is reflected diffusely (balance reflected specularly)

trick_units(-)

Definition at line 91 of file radiation_params.hh.

Referenced by jeod::RadiationFacet::define_facet_core().

8.7.4.3 thermal

```
ThermalParams jeod::RadiationParams::thermal
```

Additional thermal parameters.

trick_units(-)

Definition at line 96 of file radiation_params.hh.

Referenced by jeod::RadiationFacet::define_facet_core().

The documentation for this class was generated from the following file:

- [radiation_params.hh](#)

8.8 jeod::RadiationPressure Class Reference

Radiation pressure top-level definition.

```
#include <radiation_pressure.hh>
```

Public Member Functions

- [RadiationPressure](#) ()
- virtual [~RadiationPressure](#) ()
Destructor for [RadiationPressure](#).
- [RadiationPressure](#) (const [RadiationPressure](#) &)=delete
- [RadiationPressure](#) & operator= (const [RadiationPressure](#) &)=delete
- void [initialize](#) (DynManager &dyn_manager, [RadiationSurface](#) *surf_ptr, double center_grav[3])
Initialize the radiation pressure model when using a [RadiationSurface](#) (i.e.
- void [initialize](#) (DynManager &dyn_manager, [RadiationDefaultSurface](#) *surf_ptr)
Initialize the radiation pressure model when using a [RadiationDefaultSurface](#) (i.e.
- void [update](#) (RefFrame &vehicle_structural_frame, double center_grav[3], double scale_factor, double time)
Updates the model at each time step.
- void [add_third_body](#) ([RadiationThirdBody](#) *third_body_ptr)
Adds a third body to the vector storage of third bodies.
- void [set_third_body_active](#) (const std::string &third_body_name)
Sets a Third Body to be active when it previously was not.
- void [set_third_body_inactive](#) (const std::string &third_body_name)
Sets a Third Body to be inactive when it previously was not.
- void [set_calculate_forces](#) (bool calc_forces)
Sets the value calculate_forces.

Data Fields

- bool `active` {true}
Is radiation pressure desired?
- double `force` [3] {}
Net force due to radiation.
- double `torque` [3] {}
Net torque due to radiation.
- double `illum_factor` {}
fraction of primary flux that is not eclipsed
- `RadiationSource` `source`
Collection of data for radiation source.
- `ThermalModelRider` `thermal`
Rider to allow dynamic thermal variation on facets in this model.

Protected Member Functions

- void `update_default_surface` ()
Used to update the model when the surface is a default surface.
- void `update_facet_surface` ()
Used to update the model when the surface comprises facets.
- void `initialize_environment` (DynManager *dyn_mgr_ptr)
Initializes the source and third bodies.
- void `third_body_adjustments` (double time, RefFrame &vehicle_structural_frame)
Used to modify the flux for shadowing.
- int `find_third_body` (const std::string &third_body_name)
Finds the index of a third body based on its name.

Protected Attributes

- bool `initialized` {}
Has model been initialized?
- bool `calculate_forces` {true}
Flag to indicate whether forces and torques should be calculated.
- bool `third_bodies_active` {}
Flag indicates whether there are any of the third bodies currently active.
- unsigned int `num_third_bodies` {}
Number of ThirdBodies available.
- JeodPointerVector< `RadiationThirdBody` >::type `third_bodies`
Planetary bodies that provide shadowing or indirect, reflected, illumination.
- `RadiationSurface` * `surface_ptr` {}
The surface over which the radiation pressure will be collected.
- `RadiationDefaultSurface` * `default_surface_ptr` {}
If no surface is defined, use a DefaultSurface; this is the pointer to that default surface.
- DynManager * `dyn_manager_ptr` {}
Pointer to the Dynamics Manager.

Private Member Functions

- `JEOD_CLASS_ESTABLISH_FRIENDS2` (jeod, `RadiationPressure`)

8.8.1 Detailed Description

Radiation pressure top-level definition.

Definition at line 90 of file radiation_pressure.hh.

8.8.2 Constructor & Destructor Documentation

8.8.2.1 RadiationPressure() [1/2]

```
jeod::RadiationPressure::RadiationPressure ( )
```

Definition at line 60 of file radiation_pressure.cc.

References [third_bodies](#).

8.8.2.2 ~RadiationPressure()

```
jeod::RadiationPressure::~~RadiationPressure ( ) [virtual]
```

Destructor for [RadiationPressure](#).

Definition at line 70 of file radiation_pressure.cc.

References [third_bodies](#).

8.8.2.3 RadiationPressure() [2/2]

```
jeod::RadiationPressure::RadiationPressure (
    const RadiationPressure & ) [delete]
```

8.8.3 Member Function Documentation

8.8.3.1 add_third_body()

```
void jeod::RadiationPressure::add_third_body (
    RadiationThirdBody * third_body_ptr )
```

Adds a third body to the vector storage of third bodies.

Parameters

in	<i>third_body_ptr</i>	pointer to ThirdBody
----	-----------------------	----------------------

Definition at line 156 of file radiation_pressure.cc.

References jeod::RadiationThirdBody::active, dyn_manager_ptr, jeod::RadiationThirdBody::get_added_to_model(), jeod::RadiationMessages::incomplete_setup_error, jeod::RadiationThirdBody::initialize(), initialized, jeod::RadiationMessages::invalid_setup_error, jeod::RadiationThirdBody::name, num_third_bodies, jeod::RadiationThirdBody::set_added_to_model(), third_bodies, and third_bodies_active.

Referenced by initialize().

8.8.3.2 find_third_body()

```
int jeod::RadiationPressure::find_third_body (
    const std::string & third_body_name ) [protected]
```

Finds the index of a third body based on its name.

Returns

void

Parameters

in	<i>third_body_name</i>	Name of ThirdBody
----	------------------------	-------------------

Definition at line 358 of file radiation_pressure.cc.

References jeod::RadiationMessages::invalid_function_call, num_third_bodies, and third_bodies.

Referenced by set_third_body_active(), and set_third_body_inactive().

8.8.3.3 initialize() [1/2]

```
void jeod::RadiationPressure::initialize (
    DynManager & dyn_mgr,
    RadiationDefaultSurface * surf_ptr )
```

Initialize the radiation pressure model when using a [RadiationDefaultSurface](#) (i.e.

spherical, uniform properties.)

Parameters

in	<i>dyn_mgr</i>	The dynamics manager
in	<i>surf_ptr</i>	pointer to the radiation surface

Definition at line 69 of file radiation_pressure__default_surface.cc.

References `add_third_body()`, `default_surface_ptr`, `dyn_manager_ptr`, `jeod::RadiationDefaultSurface::initialize()`, `initialize_environment()`, `jeod::RadiationSource::num_bodies`, `jeod::RadiationSource::shadow_geometry`, `source`, `third_bodies`, and `jeod::RadiationSource::third_body`.

8.8.3.4 initialize() [2/2]

```
void jeod::RadiationPressure::initialize (
    DynManager & dyn_mgr,
    RadiationSurface * surf_ptr,
    double center_grav[3] )
```

Initialize the radiation pressure model when using a [RadiationSurface](#) (i.e. with facets).

Parameters

in	<i>dyn_mgr</i>	The dynamics manager
in	<i>surf_ptr</i>	pointer to the radiation surface
in	<i>center_grav</i>	position of center of mass Units: M

Definition at line 70 of file radiation_pressure__surface_model.cc.

References `add_third_body()`, `dyn_manager_ptr`, `jeod::RadiationSurface::initialize()`, `initialize_environment()`, `jeod::RadiationSource::num_bodies`, `jeod::RadiationSource::shadow_geometry`, `source`, `surface_ptr`, `thermal`, `third_bodies`, and `jeod::RadiationSource::third_body`.

8.8.3.5 initialize_environment()

```
void jeod::RadiationPressure::initialize_environment (
    DynManager * dyn_mgr_ptr ) [protected]
```

Initializes the source and third bodies.

Parameters

in	<i>dyn_mgr_ptr</i>	The dynamics manager
----	--------------------	----------------------

Definition at line 79 of file radiation_pressure.cc.

References `jeod::RadiationSource::initialize()`, `initialized`, `num_third_bodies`, `source`, and `third_bodies`.

Referenced by `initialize()`.

8.8.3.6 JEOD_CLASS_ESTABLISH_FRIENDS2()

```
jeod::RadiationPressure::JEOD_CLASS_ESTABLISH_FRIENDS2 (
    jeod ,
    RadiationPressure ) [private]
```

8.8.3.7 operator=()

```
RadiationPressure& jeod::RadiationPressure::operator= (
    const RadiationPressure & ) [delete]
```

8.8.3.8 set_calculate_forces()

```
void jeod::RadiationPressure::set_calculate_forces (
    bool value )
```

Sets the value calculate_forces.

Parameters

in	value	whether forces are needed
----	-------	---------------------------

Definition at line 382 of file radiation_pressure.cc.

References calculate_forces, force, and torque.

8.8.3.9 set_third_body_active()

```
void jeod::RadiationPressure::set_third_body_active (
    const std::string & third_body_name )
```

Sets a Third Body to be active when it previously was not.

Parameters

in	third_body_name	Name of ThirdBody
----	-----------------	-------------------

Definition at line 270 of file radiation_pressure.cc.

References active, dyn_manager_ptr, find_third_body(), jeod::RadiationMessages::invalid_function_call, third_bodies, and third_bodies_active.

8.8.3.10 set_third_body_inactive()

```
void jeod::RadiationPressure::set_third_body_inactive (
    const std::string & third_body_name )
```

Sets a Third Body to be inactive when it previously was not.

Parameters

in	<i>third_body_name</i>	Name of ThirdBody
----	------------------------	-------------------

Definition at line 310 of file radiation_pressure.cc.

References active, dyn_manager_ptr, find_third_body(), jeod::RadiationMessages::invalid_function_call, num_↔
third_bodies, third_bodies, and third_bodies_active.

8.8.3.11 third_body_adjustments()

```
void jeod::RadiationPressure::third_body_adjustments (
    double real_time,
    RefFrame & veh_struc_frame ) [protected]
```

Used to modify the flux for shadowing.

Parameters

in	<i>real_time</i>	The current time, e.g. TAI or UT1. Units: s
in	<i>veh_struc_frame</i>	Structure frame of vehicle

Definition at line 136 of file radiation_pressure.cc.

References active, jeod::RadiationSource::flux_inertial, jeod::RadiationSource::flux_mag, jeod::RadiationSource↔
::flux_struc, illum_factor, num_third_bodies, source, and third_bodies.

Referenced by update().

8.8.3.12 update()

```
void jeod::RadiationPressure::update (
    RefFrame & veh_struc_frame,
    double center_grav[3],
    double scale_factor,
    double real_time )
```

Updates the model at each time step.

Parameters

in	<i>veh_struc_frame</i>	The vehicle structural reference frame.
in	<i>center_grav</i>	position of center of mass Units: M
in	<i>scale_factor</i>	the scale between the simulator time and the dynamic time.
in	<i>real_time</i>	The current time, e.g. TAI or UT1. Units: s

Definition at line 97 of file radiation_pressure.cc.

References active, jeod::RadiationSource::calculate_flux(), source, surface_ptr, third_bodies_active, third_body_↵ adjustments(), update_default_surface(), and update_facet_surface().

8.8.3.13 update_default_surface()

```
void jeod::RadiationPressure::update_default_surface ( ) [protected]
```

Used to update the model when the surface is a default surface.

Definition at line 90 of file radiation_pressure__default_surface.cc.

References active, calculate_forces, default_surface_ptr, jeod::RadiationBaseFacet::F_absorption, jeod::↵ RadiationBaseFacet::F_diffuse, jeod::RadiationBaseFacet::F_specular, jeod::RadiationSource::flux_mag, jeod↵ ::RadiationSource::flux_struc_hat, force, jeod::RadiationDefaultSurface::incident_radiation(), jeod::Radiation↵ BaseFacet::initialize_runtime_values(), jeod::RadiationBaseFacet::interact_with_third_body(), num_third_bodies, source, jeod::RadiationDefaultSurface::thermal_update(), third_bodies, and torque.

Referenced by update().

8.8.3.14 update_facet_surface()

```
void jeod::RadiationPressure::update_facet_surface ( ) [protected]
```

Used to update the model when the surface comprises facets.

Definition at line 98 of file radiation_pressure__surface_model.cc.

References active, calculate_forces, jeod::RadiationSurface::equalize_absorption_emission(), jeod::Radiation↵ Source::flux_mag, jeod::RadiationSource::flux_struc_hat, force, jeod::RadiationSurface::force, jeod::Radiation↵ Surface::incident_radiation(), jeod::RadiationSurface::initialize_runtime_values(), jeod::RadiationSurface::interact↵ _with_third_body(), num_third_bodies, jeod::RadiationSurface::radiation_pressure(), source, surface_ptr, thermal, third_bodies, torque, and jeod::RadiationSurface::torque.

Referenced by update().

8.8.4 Field Documentation

8.8.4.1 active

```
bool jeod::RadiationPressure::active {true}
```

Is radiation pressure desired?

trick_units(-)

Definition at line 97 of file radiation_pressure.hh.

Referenced by set_third_body_active(), set_third_body_inactive(), third_body_adjustments(), update(), update_↵ default_surface(), and update_facet_surface().

8.8.4.2 calculate_forces

```
bool jeod::RadiationPressure::calculate_forces {true} [protected]
```

Flag to indicate whether forces and torques should be calculated.

trick_units(-)

Definition at line 133 of file radiation_pressure.hh.

Referenced by set_calculate_forces(), update_default_surface(), and update_facet_surface().

8.8.4.3 default_surface_ptr

```
RadiationDefaultSurface* jeod::RadiationPressure::default_surface_ptr {} [protected]
```

If no surface is defined, use a DefaultSurface; this is the pointer to that default surface,.

trick_units(-)

Definition at line 163 of file radiation_pressure.hh.

Referenced by initialize(), and update_default_surface().

8.8.4.4 dyn_manager_ptr

```
DynManager* jeod::RadiationPressure::dyn_manager_ptr {} [protected]
```

Pointer to the Dynamics Manager.

trick_units(-)

Definition at line 168 of file radiation_pressure.hh.

Referenced by add_third_body(), initialize(), set_third_body_active(), and set_third_body_inactive().

8.8.4.5 force

```
double jeod::RadiationPressure::force[3] {}
```

Net force due to radiation.

trick_units(N)

Definition at line 102 of file radiation_pressure.hh.

Referenced by set_calculate_forces(), update_default_surface(), and update_facet_surface().

8.8.4.6 illum_factor

```
double jeod::RadiationPressure::illum_factor {}
```

fraction of primary flux that is not eclipsed

trick_units(-)

Definition at line 112 of file radiation_pressure.hh.

Referenced by third_body_adjustments().

8.8.4.7 initialized

```
bool jeod::RadiationPressure::initialized {} [protected]
```

Has model been initialized?

trick_units(-)

Definition at line 128 of file radiation_pressure.hh.

Referenced by add_third_body(), and initialize_environment().

8.8.4.8 num_third_bodies

```
unsigned int jeod::RadiationPressure::num_third_bodies {} [protected]
```

Number of ThirdBodies available.

trick_units(count)

Definition at line 144 of file radiation_pressure.hh.

Referenced by add_third_body(), find_third_body(), initialize_environment(), set_third_body_inactive(), third_body_adjustments(), update_default_surface(), and update_facet_surface().

8.8.4.9 source

`RadiationSource` jeod::RadiationPressure::source

Collection of data for radiation source.

trick_units(-)

Definition at line 117 of file radiation_pressure.hh.

Referenced by initialize(), initialize_environment(), third_body_adjustments(), update(), update_default_surface(), and update_facet_surface().

8.8.4.10 surface_ptr

`RadiationSurface*` jeod::RadiationPressure::surface_ptr {} [protected]

The surface over which the radiation pressure will be collected.

If this is NULL, use the default radiation surface (which is a whole separate entity from a [RadiationSurface](#). trick_↔ units(-)

Definition at line 157 of file radiation_pressure.hh.

Referenced by initialize(), update(), and update_facet_surface().

8.8.4.11 thermal

`ThermalModelRider` jeod::RadiationPressure::thermal

Rider to allow dynamic thermal variation on facets in this model.

trick_units(-)

Definition at line 122 of file radiation_pressure.hh.

Referenced by initialize(), and update_facet_surface().

8.8.4.12 third_bodies

`JeodPointerVector<RadiationThirdBody>::type` jeod::RadiationPressure::third_bodies [protected]

Planetary bodies that provide shadowing or indirect, reflected, illumination.

trick_io(**)

Definition at line 150 of file radiation_pressure.hh.

Referenced by add_third_body(), find_third_body(), initialize(), initialize_environment(), RadiationPressure(), set_↔_third_body_active(), set_third_body_inactive(), third_body_adjustments(), update_default_surface(), update_↔_facet_surface(), and ~RadiationPressure().

8.8.4.13 third_bodies_active

```
bool jeod::RadiationPressure::third_bodies_active {} [protected]
```

Flag indicates whether there are any of the third bodies currently active.

trick_units(—)

Definition at line 139 of file radiation_pressure.hh.

Referenced by add_third_body(), set_third_body_active(), set_third_body_inactive(), and update().

8.8.4.14 torque

```
double jeod::RadiationPressure::torque[3] {}
```

Net torque due to radiation.

trick_units(N*m)

Definition at line 107 of file radiation_pressure.hh.

Referenced by set_calculate_forces(), update_default_surface(), and update_facet_surface().

The documentation for this class was generated from the following files:

- [radiation_pressure.hh](#)
- [radiation_pressure.cc](#)
- [radiation_pressure__default_surface.cc](#)
- [radiation_pressure__surface_model.cc](#)

8.9 jeod::RadiationSource Class Reference

Provides information on the source of the incident radiation.

```
#include <radiation_source.hh>
```

Public Types

- enum [OldShadowGeometry](#) { [Cylindrical](#) = 0 , [Cyl](#) = 1 , [Conical](#) = 2 , [Con](#) = 3 }
- distinguishes between conical and cylindrical shadowing.*

Public Member Functions

- [RadiationSource](#) ()=default
- virtual [~RadiationSource](#) ()=default
- [RadiationSource](#) (const [RadiationSource](#) &)=delete
- [RadiationSource](#) & [operator=](#) (const [RadiationSource](#) &)=delete
- virtual void [initialize](#) (DynManager *dyn_manager_ptr)
Initializes the source object for use in the Radiation Pressure model.
- virtual void [calculate_flux](#) (RefFrame &veh_struc_frame, const double center_grav[3])
calculates the flux vector from the vehicle's position.
- void [set_name](#) (std::string name_in)
Setter for the name.

Data Fields

- std::string [name](#) {"Sun"}
The name of the source of the illumination (usually Sun)
- double [flux_hat](#) [3] {}
(inertial) unit vector associated with primary radiative flux.
- double [flux_mag](#) {}
magnitude of principle radiative flux vector
- double [d_source_to_cg](#) {}
distance from source to vehicle cg.
- double [luminosity](#) {[solar_luminosity](#)}
Luminosity of primary source.
- double [radius](#) {[solar_radius](#)}
Radius of primary source.
- double [flux_inertial](#) [3] {}
(N/M/s) Power per unit area in the inertial reference frame
- double [flux_struc](#) [3] {}
*(N/M*s) Power per unit area in the vehicle structural reference frame*
- double [flux_struc_hat](#) [3] {}
Unit vector representing flux vector in the vehicle structural reference frame.
- double [inertial_cg](#) [3] {}
vehicle cg position w.r.t vehicle strucural origin, expressed in inertial RF.
- double [source_to_cg](#) [3] {}
vehicle cg position relative to the source
- double [source_to_struc_origin](#) [3] {}
vehicle struc frame origin position relative to source.
- bool [multiple_shadow_bodies](#) {}
flags that more than one body are casting shadows on the vehicle.
- RefFrame * [inertial_frame_ptr](#) {}
the inertial reference frame associated with this source
- bool [bodies_active](#) {}
can there be 3rd-body interference, either shadowing or reflection.
- unsigned int [num_bodies](#) {}
number of ThirdBodies available.
- [RadiationThirdBody](#) ** [third_body](#) {}
Planetary bodies that provide shadowing or indirect, reflected, illumination.
- [OldShadowGeometry](#) [shadow_geometry](#) {}
Flag indicating cylindrical / conical shadow geometry.

Static Public Attributes

- static constexpr double [solar_luminosity](#) {3.827E+26}
Solar Luminosity.
- static constexpr double [solar_radius](#) {6.98E+08}
Mean solar radius.

Private Member Functions

- JEOD_CLASS_ESTABLISH_FRIENDS2 [p](#) ([jeod](#), [RadiationSource](#))

8.9.1 Detailed Description

Provides information on the source of the incident radiation.

Definition at line 90 of file radiation_source.hh.

8.9.2 Member Enumeration Documentation

8.9.2.1 OldShadowGeometry

```
enum jeod::RadiationSource::OldShadowGeometry
```

distinguishes between conical and cylindrical shadowing.

NOTE this is provided for backward-compatibility and should not be used.

Enumerator

Cylindrical	planet casts a cylindrical shadow
Cyl	planet casts a cylindrical shadow
Conical	planet casts a conical shadow
Con	planet casts a conical shadow

Definition at line 200 of file radiation_source.hh.

8.9.3 Constructor & Destructor Documentation

8.9.3.1 RadiationSource() [1/2]

```
jeod::RadiationSource::RadiationSource ( ) [default]
```

8.9.3.2 ~RadiationSource()

```
virtual jeod::RadiationSource::~~RadiationSource ( ) [virtual], [default]
```

8.9.3.3 RadiationSource() [2/2]

```
jeod::RadiationSource::RadiationSource (
    const RadiationSource & ) [delete]
```

8.9.4 Member Function Documentation

8.9.4.1 calculate_flux()

```
void jeod::RadiationSource::calculate_flux (
    RefFrame & veh_struct_frame,
    const double center_gravity[3] ) [virtual]
```

calculates the flux vector from the vehicle's position.

Parameters

in	<i>veh_struct_frame</i>	the vehicle structural reference frame
in	<i>center_gravity</i>	position of the center of mass Units: M

Definition at line 72 of file radiation_source.cc.

References `d_source_to_cg`, `flux_hat`, `flux_inertial`, `flux_mag`, `flux_struct`, `flux_struct_hat`, `inertial_cg`, `inertial_`
`frame_ptr`, `luminosity`, `source_to_cg`, and `source_to_struct_origin`.

Referenced by `jeod::RadiationPressure::update()`.

8.9.4.2 initialize()

```
void jeod::RadiationSource::initialize (
    DynManager * dyn_mgr_ptr ) [virtual]
```

Initializes the source object for use in the Radiation Pressure model.

Parameters

in	<i>dyn_mgr_ptr</i>	pointer to the dynamics manager
----	--------------------	---------------------------------

Definition at line 117 of file radiation_source.cc.

References jeod::RadiationMessages::incomplete_setup_error, inertial_frame_ptr, and name.

Referenced by jeod::RadiationPressure::initialize_environment().

8.9.4.3 operator=()

```
RadiationSource& jeod::RadiationSource::operator= (
    const RadiationSource & ) [delete]
```

8.9.4.4 p()

```
JEOD_CLASS_ESTABLISH_FRIENDS2 jeod::RadiationSource::p (
    jeod ,
    RadiationSource ) [private]
```

8.9.4.5 set_name()

```
void jeod::RadiationSource::set_name (
    std::string name_in ) [inline]
```

Setter for the name.

Definition at line 227 of file radiation_source.hh.

8.9.5 Field Documentation

8.9.5.1 bodies_active

```
bool jeod::RadiationSource::bodies_active {}
```

can there be 3rd-body interference, either shadowing or reflection.

trick_units(-)

Definition at line 183 of file radiation_source.hh.

8.9.5.2 d_source_to_cg

```
double jeod::RadiationSource::d_source_to_cg {}
```

distance from source to vehicle cg.

trick_units(m)

Definition at line 121 of file radiation_source.hh.

Referenced by calculate_flux(), and jeod::RadiationThirdBody::calculate_shadow().

8.9.5.3 flux_hat

```
double jeod::RadiationSource::flux_hat[3] {}
```

(inertial) unit vector associated with primary radiative flux.

trick_units(-)

Definition at line 111 of file radiation_source.hh.

Referenced by calculate_flux().

8.9.5.4 flux_inertial

```
double jeod::RadiationSource::flux_inertial[3] {}
```

(N/M/s) Power per unit area in the inertial reference frame

trick_units(-)

Definition at line 136 of file radiation_source.hh.

Referenced by calculate_flux(), and jeod::RadiationPressure::third_body_adjustments().

8.9.5.5 flux_mag

```
double jeod::RadiationSource::flux_mag {}
```

magnitude of principle radiative flux vector

trick_units(N/m2)

Definition at line 116 of file radiation_source.hh.

Referenced by calculate_flux(), jeod::RadiationPressure::third_body_adjustments(), jeod::RadiationPressure::update_default_surface(), and jeod::RadiationPressure::update_facet_surface().

8.9.5.6 flux_struc

```
double jeod::RadiationSource::flux_struc[3] {}
```

(N/M*s) Power per unit area in the vehicle structural reference frame

trick_units(-)

Definition at line 141 of file radiation_source.hh.

Referenced by calculate_flux(), and jeod::RadiationPressure::third_body_adjustments().

8.9.5.7 flux_struc_hat

```
double jeod::RadiationSource::flux_struc_hat[3] {}
```

Unit vector representing flux vector in the vehicle structural reference frame.

trick_units(-)

Definition at line 147 of file radiation_source.hh.

Referenced by calculate_flux(), jeod::RadiationPressure::update_default_surface(), and jeod::RadiationPressure↵::update_facet_surface().

8.9.5.8 inertial_cg

```
double jeod::RadiationSource::inertial_cg[3] {}
```

vehicle cg position w.r.t vehicle structural origin, expressed in inertial RF.

trick_units(m)

Definition at line 153 of file radiation_source.hh.

Referenced by calculate_flux().

8.9.5.9 inertial_frame_ptr

```
RefFrame* jeod::RadiationSource::inertial_frame_ptr {}
```

the inertial reference frame associated with this source

trick_units(-)

Definition at line 173 of file radiation_source.hh.

Referenced by calculate_flux(), initialize(), jeod::RadiationThirdBody::initialize(), and jeod::RadiationThirdBody↵::update_third_body_state().

8.9.5.10 luminosity

```
double jeod::RadiationSource::luminosity {solar_luminosity}
```

Luminosity of primary source.

trick_units(-)

Definition at line 126 of file radiation_source.hh.

Referenced by calculate_flux().

8.9.5.11 multiple_shadow_bodies

```
bool jeod::RadiationSource::multiple_shadow_bodies {}
```

flags that more than one body are casting shadows on the vehicle.

trick_units(-)

Definition at line 168 of file radiation_source.hh.

8.9.5.12 name

```
std::string jeod::RadiationSource::name {"Sun"}
```

The name of the source of the illumination (usually Sun)

trick_units(-)

Definition at line 106 of file radiation_source.hh.

Referenced by initialize().

8.9.5.13 num_bodies

```
unsigned int jeod::RadiationSource::num_bodies {}
```

number of ThirdBodies available.

trick_units(count)

Definition at line 188 of file radiation_source.hh.

Referenced by jeod::RadiationPressure::initialize().

8.9.5.14 radius

```
double jeod::RadiationSource::radius {solar_radius}
```

Radius of primary source.

trick_units(m)

Definition at line 131 of file radiation_source.hh.

Referenced by jeod::RadiationThirdBody::calculate_shadow(), and jeod::RadiationThirdBody::initialize().

8.9.5.15 shadow_geometry

```
OldShadowGeometry jeod::RadiationSource::shadow_geometry {}
```

Flag indicating cylindrical / conical shadow geometry.

trick_units(-)

Definition at line 211 of file radiation_source.hh.

Referenced by jeod::RadiationPressure::initialize().

8.9.5.16 solar_luminosity

```
constexpr double jeod::RadiationSource::solar_luminosity {3.827E+26} [static], [constexpr]
```

Solar Luminosity.

trick_units(-)

Definition at line 96 of file radiation_source.hh.

8.9.5.17 solar_radius

```
constexpr double jeod::RadiationSource::solar_radius {6.98E+08} [static], [constexpr]
```

Mean solar radius.

trick_units(m)

Definition at line 101 of file radiation_source.hh.

8.9.5.18 source_to_cg

```
double jeod::RadiationSource::source_to_cg[3] {}
```

vehicle cg position relative to the source

trick_units(m)

Definition at line 158 of file radiation_source.hh.

Referenced by calculate_flux(), and jeod::RadiationThirdBody::calculate_shadow().

8.9.5.19 source_to_struc_origin

```
double jeod::RadiationSource::source_to_struc_origin[3] {}
```

vehicle struc frame origin position relative to source.

trick_units(m)

Definition at line 163 of file radiation_source.hh.

Referenced by calculate_flux().

8.9.5.20 third_body

```
RadiationThirdBody** jeod::RadiationSource::third_body {}
```

Planetary bodies that provide shadowing or indirect, reflected, illumination.

trick_units(-)

Definition at line 194 of file radiation_source.hh.

Referenced by jeod::RadiationPressure::initialize().

The documentation for this class was generated from the following files:

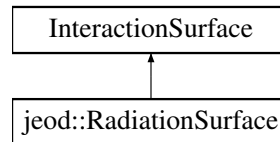
- [radiation_source.hh](#)
- [radiation_source.cc](#)

8.10 jeod::RadiationSurface Class Reference

The surface of the vehicle that interacts with the incident flux.

```
#include <radiation_surface.hh>
```

Inheritance diagram for jeod::RadiationSurface:



Public Member Functions

- [RadiationSurface](#) ()
Constructor for [RadiationSurface](#).
- [~RadiationSurface](#) () override
Destructor for [RadiationSurface](#).
- [RadiationSurface](#) & [operator=](#) (const [RadiationSurface](#) &)=delete
- [RadiationSurface](#) (const [RadiationSurface](#) &)=delete
- void [initialize](#) (double center_grav[3])
Initializes the radiation surface.
- void [allocate_array](#) (unsigned int size) override
Allocates memory for an array of radiation facets.
- void [allocate_interaction_facet](#) (Facet *facet, InteractionFacetFactory *factory, FacetParams *params, unsigned int index) override
Turns facet memory into radiation facet memory.
- void [initialize_runtime_values](#) ()
To initialize the values during each update run.
- void [incident_radiation](#) (double flux_mag, const double flux_struc_hat[3], bool calculate_forces)
systematically calls the method to calculate the interaction on each facet.
- void [interact_with_third_body](#) ([RadiationThirdBody](#) *third_body_ptr, const bool calculate_forces)
systematically calls the method to calculate the interaction of each facet with ThirdBody flux.
- void [accumulate_thermal_sources](#) () override
systematically calls the method to accumulate thermal sources on each facet.
- void [thermal_integrator](#) () override
systematically calls the method to integrate the temperature variation for each facet.
- void [equalize_absorption_emission](#) ()
systematically calls the method to ensure that the same for each facet.
- void [radiation_pressure](#) ()
systematically calls the method to wrap up the radiation pressure calculation on each facet and accumulate forces and torques over all facets.
- void [add_thermal_integrators_to](#) (DynBody *dyn_body)
To attach thermal integrators from each facet to the integration group of a DynBody.

Data Fields

- [RadiationFacet](#) ** [facets](#) {}
Array of pointers to the Radiation Facets.
- unsigned int [num_facets](#) {}
number of facets in this surface.
- bool [include_conduction](#) {}
Flag to instruct model to include conduction between plates in the thermal calculation.
- double ** [thermal_conduction](#) {}
Conductivity values (Watts per Kelvin) between facets.
- double [force](#) [3] {}
Force resulting from all radiative interactions.
- double [torque](#) [3] {}
Torque resulting from all radiative interactions.
- unsigned int [ii_facet](#) {}
Simple counter, used repeatedly.

Private Member Functions

- JEOD_CLASS_ESTABLISH_FRIENDS2 [p](#) (jeod, [RadiationSurface](#))

8.10.1 Detailed Description

The surface of the vehicle that interacts with the incident flux.

Definition at line 88 of file radiation_surface.hh.

8.10.2 Constructor & Destructor Documentation

8.10.2.1 RadiationSurface() [1/2]

```
jeod::RadiationSurface::RadiationSurface ( )
```

Constructor for [RadiationSurface](#).

Definition at line 60 of file radiation_surface.cc.

8.10.2.2 ~RadiationSurface()

```
jeod::RadiationSurface::~~RadiationSurface ( ) [override]
```

Destructor for [RadiationSurface](#).

Definition at line 337 of file radiation_surface.cc.

References [facets](#), and [num_facets](#).

8.10.2.3 RadiationSurface() [2/2]

```
jeod::RadiationSurface::RadiationSurface (
    const RadiationSurface & ) [delete]
```

8.10.3 Member Function Documentation

8.10.3.1 accumulate_thermal_sources()

```
void jeod::RadiationSurface::accumulate_thermal_sources ( ) [override]
```

systematically calls the method to accumulate thermal sources on each facet.

Definition at line 255 of file radiation_surface.cc.

References facets, ii_facet, num_facets, and jeod::RadiationBaseFacet::thermal.

8.10.3.2 add_thermal_integrators_to()

```
void jeod::RadiationSurface::add_thermal_integrators_to (
    DynBody * dyn_body )
```

To attach thermal integrators from each facet to the integration group of a DynBody.

Parameters

in, out	<i>dyn_body</i>	Body to which integrators are to be added.
---------	-----------------	--

Definition at line 325 of file radiation_surface.cc.

References facets, num_facets, and jeod::RadiationBaseFacet::thermal.

8.10.3.3 allocate_array()

```
void jeod::RadiationSurface::allocate_array (
    unsigned int size ) [override]
```

Allocates memory for an array of radiation facets.

Parameters

in	<i>size</i>	Size of array
----	-------------	---------------

Definition at line 134 of file radiation_surface.cc.

References facets, ii_facet, num_facets, and jeod::RadiationMessages::operational_setup_error.

8.10.3.4 allocate_interaction_facet()

```
void jeod::RadiationSurface::allocate_interaction_facet (
    Facet * facet,
    InteractionFacetFactory * factory,
    FacetParams * params,
    unsigned int index ) [override]
```

Turns facet memory into radiaiton facet memory.

Parameters

in	<i>facet</i>	pointer to the facet
in	<i>factory</i>	pointer to the facet factory
in	<i>params</i>	pointer to the generic facet parameter.
in	<i>index</i>	index value in the facet list.

Definition at line 166 of file radiation_surface.cc.

References facets, jeod::RadiationMessages::invalid_setup_error, num_facets, and jeod::RadiationMessages::operational_setup_error.

8.10.3.5 equalize_absorption_emission()

```
void jeod::RadiationSurface::equalize_absorption_emission ( )
```

systematically calls the method to ensure that the same for each facet.

Definition at line 284 of file radiation_surface.cc.

References facets, ii_facet, num_facets, and jeod::RadiationBaseFacet::thermal.

Referenced by jeod::RadiationPressure::update_facet_surface().

8.10.3.6 incident_radiation()

```
void jeod::RadiationSurface::incident_radiation (
    double flux_mag,
    const double flux_struct_hat[3],
    bool calculate_forces )
```

systematically calls the method to calculate the interaction on each facet.

Parameters

in	<i>flux_mag</i>	Magnitude of incident flux
in	<i>flux_struc_hat</i>	unit vector of incident flux
in	<i>calculate_forces</i>	boolean indicating whether to calculate forces.

Definition at line 230 of file radiation_surface.cc.

References facets, ii_facet, jeod::RadiationBaseFacet::incident_radiation(), and num_facets.

Referenced by jeod::RadiationPressure::update_facet_surface().

8.10.3.7 initialize()

```
void jeod::RadiationSurface::initialize (
    double center_grav[3] )
```

Initializes the radiation surface.

Parameters

in	<i>center_grav</i>	position of center of mass Units: M
----	--------------------	--

Definition at line 69 of file radiation_surface.cc.

References facets, ii_facet, jeod::RadiationFacet::initialize_geom(), jeod::RadiationMessages::invalid_setup_error, num_facets, jeod::RadiationMessages::operational_setup_error, and jeod::RadiationBaseFacet::thermal.

Referenced by jeod::RadiationPressure::initialize().

8.10.3.8 initialize_runtime_values()

```
void jeod::RadiationSurface::initialize_runtime_values ( )
```

To initialize the values during each update run.

Definition at line 312 of file radiation_surface.cc.

References facets, ii_facet, jeod::RadiationBaseFacet::initialize_runtime_values(), and num_facets.

Referenced by jeod::RadiationPressure::update_facet_surface().

8.10.3.9 interact_with_third_body()

```
void jeod::RadiationSurface::interact_with_third_body (
    RadiationThirdBody * third_body_ptr,
    const bool calculate_forces )
```

systematically calls the method to calculate the interaction of each facet with ThirdBody flux.

Parameters

in, out	<i>third_body_ptr</i>	Third body that emits radiation.
in	<i>calculate_forces</i>	Calculate forces on the surface if true.

Definition at line 244 of file radiation_surface.cc.

References facets, ii_facet, jeod::RadiationBaseFacet::interact_with_third_body(), and num_facets.

Referenced by jeod::RadiationPressure::update_facet_surface().

8.10.3.10 operator=()

```
RadiationSurface& jeod::RadiationSurface::operator= (
    const RadiationSurface & ) [delete]
```

8.10.3.11 p()

```
JEOD_CLASS_ESTABLISH_FRIENDS2 jeod::RadiationSurface::p (
    jeod ,
    RadiationSurface ) [private]
```

8.10.3.12 radiation_pressure()

```
void jeod::RadiationSurface::radiation_pressure ( )
```

systematically calls the method to wrap up the radiation pressure calculation on each facet and accumulate forces and torques over all facets.

Definition at line 296 of file radiation_surface.cc.

References facets, force, ii_facet, num_facets, jeod::RadiationFacet::radiation_pressure(), and torque.

Referenced by jeod::RadiationPressure::update_facet_surface().

8.10.3.13 thermal_integrator()

```
void jeod::RadiationSurface::thermal_integrator ( ) [override]
```

systematically calls the method to integrate the temperature variation for each facet.

Definition at line 266 of file radiation_surface.cc.

References facets, ii_facet, num_facets, and jeod::RadiationBaseFacet::thermal.

8.10.4 Field Documentation

8.10.4.1 facets

```
RadiationFacet** jeod::RadiationSurface::facets {}
```

Array of pointers to the Radiation Facets.

trick_units(-)

Definition at line 94 of file radiation_surface.hh.

Referenced by accumulate_thermal_sources(), add_thermal_integrators_to(), allocate_array(), allocate_↵ interaction_facet(), equalize_absorption_emission(), incident_radiation(), initialize(), initialize_runtime_values(), interact_with_third_body(), radiation_pressure(), thermal_integrator(), and ~RadiationSurface().

8.10.4.2 force

```
double jeod::RadiationSurface::force[3] {}
```

Force resulting from all radiative interactions.

trick_units(-)

Definition at line 118 of file radiation_surface.hh.

Referenced by radiation_pressure(), and jeod::RadiationPressure::update_facet_surface().

8.10.4.3 ii_facet

```
unsigned int jeod::RadiationSurface::ii_facet {}
```

Simple counter, used repeatedly.

trick_units(-)

Definition at line 128 of file radiation_surface.hh.

Referenced by accumulate_thermal_sources(), allocate_array(), equalize_absorption_emission(), incident_↵ radiation(), initialize(), initialize_runtime_values(), interact_with_third_body(), radiation_pressure(), and thermal_↵ integrator().

8.10.4.4 include_conduction

```
bool jeod::RadiationSurface::include_conduction {}
```

Flag to instruct model to include conduction between plates in the thermal calculation.

Requires a thermal_conduction matrix. trick_units(-)

Definition at line 105 of file radiation_surface.hh.

8.10.4.5 num_facets

```
unsigned int jeod::RadiationSurface::num_facets {}
```

number of facets in this surface.

trick_units(-)

Definition at line 99 of file radiation_surface.hh.

Referenced by accumulate_thermal_sources(), add_thermal_integrators_to(), allocate_array(), allocate_interaction_facet(), equalize_absorption_emission(), incident_radiation(), initialize(), initialize_runtime_values(), interact_with_third_body(), radiation_pressure(), thermal_integrator(), and ~RadiationSurface().

8.10.4.6 thermal_conduction

```
double** jeod::RadiationSurface::thermal_conduction {}
```

Conductivity values (Watts per Kelvin) between facets.

Assumed symmetric, only the values [small][large] will be considered (e.g. thermal_conduction[2][3] gives the conduction between facets 2 and 3, whereas thermal_conduction[3][2] will never be seen. trick_units(-)

Definition at line 113 of file radiation_surface.hh.

8.10.4.7 torque

```
double jeod::RadiationSurface::torque[3] {}
```

Torque resulting from all radiative interactions.

trick_units(-)

Definition at line 123 of file radiation_surface.hh.

Referenced by radiation_pressure(), and jeod::RadiationPressure::update_facet_surface().

The documentation for this class was generated from the following files:

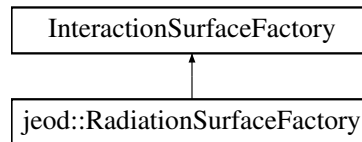
- [radiation_surface.hh](#)
- [radiation_surface.cc](#)

8.11 jeod::RadiationSurfaceFactory Class Reference

The factory for creating Radiation Surfaces.

```
#include <radiation_surface_factory.hh>
```

Inheritance diagram for jeod::RadiationSurfaceFactory:



Public Member Functions

- [RadiationSurfaceFactory](#) ()
Constructor.
- [~RadiationSurfaceFactory](#) () override
Destructor.
- [RadiationSurfaceFactory](#) & [operator=](#) (const [RadiationSurfaceFactory](#) &)=delete
- [RadiationSurfaceFactory](#) (const [RadiationSurfaceFactory](#) &)=delete
- void [add_facet_params](#) (FacetParams *to_add) override
Add a set of radiation parameters to those available to Radiation Surface.

Protected Attributes

- [FlatPlateRadiationFactory](#) [flat_plate_radiation_factory](#)
The factory to build flat plate thermal facets.

Private Member Functions

- JEOD_CLASS_ESTABLISH_FRIENDS2 [p](#) (jeod, [RadiationSurfaceFactory](#))

8.11.1 Detailed Description

The factory for creating Radiation Surfaces.

Definition at line 85 of file radiation_surface_factory.hh.

8.11.2 Constructor & Destructor Documentation

8.11.2.1 RadiationSurfaceFactory() [1/2]

```
jeod::RadiationSurfaceFactory::RadiationSurfaceFactory ( )
```

Constructor.

Definition at line 52 of file radiation_surface_factory.cc.

References flat_plate_radiation_factory.

8.11.2.2 ~RadiationSurfaceFactory()

```
jeod::RadiationSurfaceFactory::~~RadiationSurfaceFactory ( ) [override]
```

Destructor.

Definition at line 91 of file radiation_surface_factory.cc.

8.11.2.3 RadiationSurfaceFactory() [2/2]

```
jeod::RadiationSurfaceFactory::RadiationSurfaceFactory (
    const RadiationSurfaceFactory & ) [delete]
```

8.11.3 Member Function Documentation

8.11.3.1 add_facet_params()

```
void jeod::RadiationSurfaceFactory::add_facet_params (
    FacetParams * to_add ) [override]
```

Add a set of radiation parameters to those available to Radiation Surface.

Parameters

in	to_add	pointer to the parameter list to add.
----	--------	---------------------------------------

Definition at line 64 of file radiation_surface_factory.cc.

References jeod::RadiationMessages::invalid_setup_error.

8.11.3.2 operator=()

```
RadiationSurfaceFactory& jeod::RadiationSurfaceFactory::operator= (
    const RadiationSurfaceFactory & ) [delete]
```

8.11.3.3 p()

```
JEOD_CLASS_ESTABLISH_FRIENDS2 jeod::RadiationSurfaceFactory::p (
    jeod ,
    RadiationSurfaceFactory ) [private]
```

8.11.4 Field Documentation

8.11.4.1 flat_plate_radiation_factory

```
FlatPlateRadiationFactory jeod::RadiationSurfaceFactory::flat_plate_radiation_factory [protected]
```

The factory to build flat plate thermal facets.

trick_units(-)

Definition at line 103 of file radiation_surface_factory.hh.

Referenced by RadiationSurfaceFactory().

The documentation for this class was generated from the following files:

- [radiation_surface_factory.hh](#)
- [radiation_surface_factory.cc](#)

8.12 jeod::RadiationThirdBody Class Reference

Provide information on bodies that may cause shadowing or reflected illumination.

```
#include <radiation_third_body.hh>
```

Public Types

- enum [ShadowGeometry](#) { [Cylindrical](#) = 0 , [Cyl](#) = 1 , [Conical](#) = 2 , [Con](#) = 3 }
distinguishes between conical and cylindrical shadowing

Public Member Functions

- [RadiationThirdBody](#) ()=default
- virtual [~RadiationThirdBody](#) ()=default
- [RadiationThirdBody](#) (const [RadiationThirdBody](#) &)=delete
- [RadiationThirdBody](#) & operator= (const [RadiationThirdBody](#) &)=delete
- virtual void [initialize](#) (DynManager *dyn_mgr_ptr)
initializes the third body
- void [convert_shadow_from_int](#) (int geometry)
convert from the old style framework in which the shadow geometry was owned by the [RadiationSource](#) (same for all third bodies) to the new style framework where the geometry is owned by each [ThirdBody](#)
- virtual double [process_third_body](#) (double real_time, RefFrame &veh_struc_frame)
Primary executable for handling the Third Body effects.
- void [set_name](#) (std::string name_in)
Setter for the name.
- virtual void [accumulate_refl_flux](#) ([RadiationBaseFacet](#) *veh_surf_elem, bool calculate_forces)
To provide base class null implementation.
- virtual void [accumulate_rad_flux](#) ([RadiationBaseFacet](#) *veh_surf_elem, bool calculate_forces)
To provide base class null implementation.
- virtual bool [is_interactive](#) ()
Identifies this class as one that does not produce a radiation field.
- bool [get_added_to_model](#) ()
Returns information on whether the body has already been used in one [RadiationPressure](#) model.
- void [set_added_to_model](#) (bool value)
Sets the added_to_model value.

Data Fields

- [RadiationSource](#) * [primary_source_ptr](#) {}
Pointer to the primary illumination source.
- [ShadowGeometry](#) [shadow_geometry](#) {Conical}
Flag indicating cylindrical / conical shadow geometry.
- bool [force_state_update](#) {true}
Flag to indicate whether to enforce an update to the third-body body-state.
- std::string [name](#)
Name of planet.
- bool [active](#) {true}
Model on/off flag.
- double [update_interval](#) {}
Time interval between third body state updates.
- double [radius](#) {}
Radius of [RadiationThirdBody](#).
- double [output_flux](#) [3] {}
Calculated value.
- RefFrame * [local_frame_ptr](#) {}
Pointer to the local reference frame, for determining position of this object.

Protected Member Functions

- double [generate_alpha](#) (double rho_adj, double delta)
generate_alpha(rho_adj, delta: generates an approximate alpha-value given a relative disk size and fraction of maximum possible eclipse
- bool [test_for_state_update](#) (double time)
Tests for necessity of updating third body state, and calls appropriate update method (polymorphic) if needed.
- void [calculate_shadow](#) ()
Calculates the effect of shadowing by a third body.
- virtual bool [update_third_body_state](#) ()
Updates the state of the Third Body.

Protected Attributes

- bool [added_to_model](#) {}
Flag to indicate whether body has already been added to a model.
- bool [initialized](#) {}
Flag to indicate that initialization has been complete.
- double [illum_factor](#) {1.0}
Value between 0 and 1 representing the fraction of possible flux that is actually incident after shadowing effects.
- double [last_update_time](#) {}
Time at which the third body state was last updated.
- Planet * [planet_link](#) {}
Pointer to the associated Planet object.
- double [r_plus](#) {}
The value (radius + R_source)
- double [r_minus](#) {}
The value (radius - R_source)
- double [r_ratio](#) {}
The ratio (radius / R_source)
- double [r_par](#) {}
Component of vector from [RadiationThirdBody](#) to vehicle that is parallel to vector from primary source to [RadiationThirdBody](#).
- double [r_perp](#) {}
Component of vector from [RadiationThirdBody](#) to vehicle that is perpendicular to vector from primary source to [RadiationThirdBody](#).
- double [third_to_cg_inrtl](#) [3] {}
Vector from [RadiationThirdBody](#) to vehicle center of gravity, in inertial RF.
- double [source_to_third_inrtl](#) [3] {}
Vector from the primary source to the [RadiationThirdBody](#) in inertial RF.
- double [d_source_to_third](#) {}
Magnitude of source_to_third.
- double [source_to_third_hat_inrtl](#) [3] {}
Unit vector of source_to_third.

Private Member Functions

- JEOD_CLASS_ESTABLISH_FRIENDS2 p (jeod, [RadiationThirdBody](#))

8.12.1 Detailed Description

Provide information on bodies that may cause shadowing or reflected illumination.

Definition at line 84 of file radiation_third_body.hh.

8.12.2 Member Enumeration Documentation

8.12.2.1 ShadowGeometry

```
enum jeod::RadiationThirdBody::ShadowGeometry
```

distinguishes between conical and cylindrical shadowing

Enumerator

Cylindrical	planet casts a cylindrical shadow
Cyl	planet casts a cylindrical shadow
Conical	planet casts a conical shadow
Con	planet casts a conical shadow

Definition at line 90 of file radiation_third_body.hh.

8.12.3 Constructor & Destructor Documentation

8.12.3.1 RadiationThirdBody() [1/2]

```
jeod::RadiationThirdBody::RadiationThirdBody ( ) [default]
```

8.12.3.2 ~RadiationThirdBody()

```
virtual jeod::RadiationThirdBody::~~RadiationThirdBody ( ) [virtual], [default]
```

8.12.3.3 RadiationThirdBody() [2/2]

```
jeod::RadiationThirdBody::RadiationThirdBody (
    const RadiationThirdBody & ) [delete]
```

8.12.4 Member Function Documentation

8.12.4.1 accumulate_rad_flux()

```
virtual void jeod::RadiationThirdBody::accumulate_rad_flux (
    RadiationBaseFacet * veh_surf_elem,
    bool calculate_forces ) [inline], [virtual]
```

To provide base class null implementation.

Parameters

<i>veh_surf_elem</i>	The facet of the vehicle.
<i>calculate_forces</i>	pass-through flag.

Definition at line 262 of file radiation_third_body.hh.

Referenced by jeod::RadiationBaseFacet::interact_with_third_body().

8.12.4.2 accumulate_refl_flux()

```
virtual void jeod::RadiationThirdBody::accumulate_refl_flux (
    RadiationBaseFacet * veh_surf_elem,
    bool calculate_forces ) [inline], [virtual]
```

To provide base class null implementation.

Parameters

<i>veh_surf_elem</i>	The facet of the vehicle.
<i>calculate_forces</i>	pass-through flag.

Definition at line 253 of file radiation_third_body.hh.

Referenced by jeod::RadiationBaseFacet::interact_with_third_body().

8.12.4.3 calculate_shadow()

```
void jeod::RadiationThirdBody::calculate_shadow ( ) [protected]
```

Calculates the effect of shadowing by a third body.

Definition at line 190 of file radiation_third_body.cc.

References `Con`, `Conical`, `Cyl`, `Cylindrical`, `jeod::RadiationSource::d_source_to_cg`, `d_source_to_third`, `generate_alpha()`, `illum_factor`, `jeod::RadiationMessages::invalid_setup_error`, `name`, `primary_source_ptr`, `r_minus`, `r_par`, `r_perp`, `r_plus`, `r_ratio`, `jeod::RadiationSource::radius`, `radius`, `shadow_geometry`, `jeod::RadiationSource::source_to_cg`, `source_to_third_hat_inrtl`, `source_to_third_inrtl`, and `third_to_cg_inrtl`.

Referenced by `process_third_body()`.

8.12.4.4 convert_shadow_from_int()

```
void jeod::RadiationThirdBody::convert_shadow_from_int (
    int old_shadow_geometry )
```

convert from the old style framework in which the shadow geometry was owned by the [RadiationSource](#) (same for all third bodies) to the new style framework where the geometry is owned by each `ThirdBody`

Parameters

in	<i>old_shadow_geometry</i>	integer representation of the shadow geometry enumeration.
----	----------------------------	--

Definition at line 350 of file `radiation_third_body.cc`.

References `jeod::RadiationMessages::invalid_setup_error`, and `shadow_geometry`.

8.12.4.5 generate_alpha()

```
double jeod::RadiationThirdBody::generate_alpha (
    double rho_adj,
    double delta ) [protected]
```

`generate_alpha( rho_adj, delta`: generates an approximate alpha-value given a relative disk size and fraction of maximum possible eclipse

Parameters

in	<i>rho_adj</i>	relative disk size
in	<i>delta</i>	fraction of eclipse achieved.

Definition at line 331 of file `radiation_third_body.cc`.

Referenced by `calculate_shadow()`.

8.12.4.6 get_added_to_model()

```
bool jeod::RadiationThirdBody::get_added_to_model ( ) [inline]
```

Returns information on whether the body has already been used in one [RadiationPressure](#) model.

Protects against multiple use of one `ThirdBody` instance.

Returns

added_to_model

Definition at line 281 of file radiation_third_body.hh.

Referenced by jeod::RadiationPressure::add_third_body().

8.12.4.7 initialize()

```
void jeod::RadiationThirdBody::initialize (
    DynManager * dyn_mgr_ptr ) [virtual]
```

initializes the third body

Parameters

in	<i>dyn_mgr_ptr</i>	pointer to the dynamics manager
----	--------------------	---------------------------------

Definition at line 57 of file radiation_third_body.cc.

References active, jeod::RadiationSource::inertial_frame_ptr, initialized, jeod::RadiationMessages::invalid_setup↵_error, local_frame_ptr, name, planet_link, primary_source_ptr, r_minus, r_plus, r_ratio, jeod::RadiationSource↵::radius, and radius.

Referenced by jeod::RadiationPressure::add_third_body().

8.12.4.8 is_interactive()

```
virtual bool jeod::RadiationThirdBody::is_interactive ( ) [inline], [virtual]
```

Identifies this class as one that does not produce a radiation field.

Returns

false

Definition at line 270 of file radiation_third_body.hh.

8.12.4.9 operator=()

```
RadiationThirdBody& jeod::RadiationThirdBody::operator= (
    const RadiationThirdBody & ) [delete]
```

8.12.4.10 p()

```
JEOD_CLASS_ESTABLISH_FRIENDS2 jeod::RadiationThirdBody::p (
    jeod ,
    RadiationThirdBody ) [private]
```

8.12.4.11 process_third_body()

```
double jeod::RadiationThirdBody::process_third_body (
    double real_time,
    RefFrame & veh_struc_frame ) [virtual]
```

Primary executable for handling the Third Body effects.

Returns

shading fraction

Parameters

in	<i>real_time</i>	Current time. Units: s
in	<i>veh_struc_frame</i>	Unused.

Definition at line 371 of file radiation_third_body.cc.

References `active`, `calculate_shadow()`, `illum_factor`, `initialized`, `jeod::RadiationMessages::invalid_setup_error`, `name`, `jeod::RadiationMessages::operational_setup_error`, and `test_for_state_update()`.

8.12.4.12 set_added_to_model()

```
void jeod::RadiationThirdBody::set_added_to_model (
    bool value ) [inline]
```

Sets the `added_to_model` value.

Parameters

<i>value</i>	New value
--------------	-----------

Definition at line 290 of file radiation_third_body.hh.

Referenced by `jeod::RadiationPressure::add_third_body()`.

8.12.4.13 set_name()

```
void jeod::RadiationThirdBody::set_name (
    std::string name_in ) [inline]
```

Setter for the name.

Definition at line 243 of file radiation_third_body.hh.

8.12.4.14 test_for_state_update()

```
bool jeod::RadiationThirdBody::test_for_state_update (
    double real_time ) [protected]
```

Tests for necessity of updating third body state, and calls appropriate update method (polymorphic) if needed.

Parameters

in	<i>real_time</i>	Current time Units: s
----	------------------	--------------------------

Definition at line 418 of file radiation_third_body.cc.

References force_state_update, last_update_time, update_interval, and update_third_body_state().

Referenced by process_third_body().

8.12.4.15 update_third_body_state()

```
bool jeod::RadiationThirdBody::update_third_body_state ( ) [protected], [virtual]
```

Updates the state of the Third Body.

Because the ThirdBody may be associated with multiple vehicles, and because its state typically changes much more slowly than the vehicle anyway, it is not always necessary to update the ThirdBodyState.

Definition at line 440 of file radiation_third_body.cc.

References active, d_source_to_third, jeod::RadiationSource::inertial_frame_ptr, jeod::RadiationMessages↔::invalid_setup_error, local_frame_ptr, name, primary_source_ptr, source_to_third_hat_inrtl, and source_to_↔third_inrtl.

Referenced by test_for_state_update().

8.12.5 Field Documentation

8.12.5.1 active

```
bool jeod::RadiationThirdBody::active {true}
```

Model on/off flag.

trick_units(-)

Definition at line 123 of file radiation_third_body.hh.

Referenced by jeod::RadiationPressure::add_third_body(), initialize(), process_third_body(), and update_third_body_state().

8.12.5.2 added_to_model

```
bool jeod::RadiationThirdBody::added_to_model {} [protected]
```

Flag to indicate whether body has already been added to a model.

trick_units(-)

Definition at line 153 of file radiation_third_body.hh.

8.12.5.3 d_source_to_third

```
double jeod::RadiationThirdBody::d_source_to_third {} [protected]
```

Magnitude of source_to_third.

trick_units(m)

Definition at line 220 of file radiation_third_body.hh.

Referenced by calculate_shadow(), and update_third_body_state().

8.12.5.4 force_state_update

```
bool jeod::RadiationThirdBody::force_state_update {true}
```

Flag to indicate whether to enforce an update to the third-body body-state.

Default: true Self-resets to false after each use. trick_units(-)

Definition at line 113 of file radiation_third_body.hh.

Referenced by test_for_state_update().

8.12.5.5 illum_factor

```
double jeod::RadiationThirdBody::illum_factor {1.0} [protected]
```

Value between 0 and 1 representing the fraction of possible flux that is actually incident after shadowing effects.

trick_units(-)

Definition at line 164 of file radiation_third_body.hh.

Referenced by calculate_shadow(), and process_third_body().

8.12.5.6 initialized

```
bool jeod::RadiationThirdBody::initialized {} [protected]
```

Flag to indicate that initialization has been complete.

trick_units(-)

Definition at line 158 of file radiation_third_body.hh.

Referenced by initialize(), and process_third_body().

8.12.5.7 last_update_time

```
double jeod::RadiationThirdBody::last_update_time {} [protected]
```

Time at which the third body state was last updated.

trick_units(s)

Definition at line 169 of file radiation_third_body.hh.

Referenced by test_for_state_update().

8.12.5.8 local_frame_ptr

```
RefFrame* jeod::RadiationThirdBody::local_frame_ptr {}
```

Pointer to the local reference frame, for determining position of this object.

trick_units(-)

Definition at line 146 of file radiation_third_body.hh.

Referenced by initialize(), and update_third_body_state().

8.12.5.9 name

```
std::string jeod::RadiationThirdBody::name
```

Name of planet.

trick_units(-)

Definition at line 118 of file radiation_third_body.hh.

Referenced by jeod::RadiationPressure::add_third_body(), calculate_shadow(), initialize(), process_third_body(), and update_third_body_state().

8.12.5.10 output_flux

```
double jeod::RadiationThirdBody::output_flux[3] {}
```

Calculated value.

Set only in derived class RadiationReflectingThirdBody. trick_units(-)

Definition at line 140 of file radiation_third_body.hh.

8.12.5.11 planet_link

```
Planet* jeod::RadiationThirdBody::planet_link {} [protected]
```

Pointer to the associated Planet object.

trick_units(-)

Definition at line 175 of file radiation_third_body.hh.

Referenced by initialize().

8.12.5.12 primary_source_ptr

```
RadiationSource* jeod::RadiationThirdBody::primary_source_ptr {}
```

Pointer to the primary illumination source.

trick_units(-)

Definition at line 101 of file radiation_third_body.hh.

Referenced by calculate_shadow(), initialize(), and update_third_body_state().

8.12.5.13 r_minus

```
double jeod::RadiationThirdBody::r_minus {} [protected]
```

The value (radius - R_source)

trick_units(-)

Definition at line 185 of file radiation_third_body.hh.

Referenced by calculate_shadow(), and initialize().

8.12.5.14 r_par

```
double jeod::RadiationThirdBody::r_par {} [protected]
```

Component of vector from [RadiationThirdBody](#) to vehicle that is parallel to vector from primary source to [RadiationThirdBody](#).

trick_units(m)

Definition at line 197 of file radiation_third_body.hh.

Referenced by calculate_shadow().

8.12.5.15 r_perp

```
double jeod::RadiationThirdBody::r_perp {} [protected]
```

Component of vector from [RadiationThirdBody](#) to vehicle that is perpendicular to vector from primary source to [RadiationThirdBody](#).

trick_units(-)

Definition at line 203 of file radiation_third_body.hh.

Referenced by calculate_shadow().

8.12.5.16 r_plus

```
double jeod::RadiationThirdBody::r_plus {} [protected]
```

The value (radius + R_source)

trick_units(-)

Definition at line 180 of file radiation_third_body.hh.

Referenced by calculate_shadow(), and initialize().

8.12.5.17 r_ratio

```
double jeod::RadiationThirdBody::r_ratio {} [protected]
```

The ratio (radius / R_source)

trick_units(-)

Definition at line 190 of file radiation_third_body.hh.

Referenced by calculate_shadow(), and initialize().

8.12.5.18 radius

```
double jeod::RadiationThirdBody::radius {}
```

Radius of [RadiationThirdBody](#).

Note - this is set automatically if RTB is a planet, but must be set manually if RTB is a DynBody. trick_units(m)

Definition at line 134 of file radiation_third_body.hh.

Referenced by calculate_shadow(), and initialize().

8.12.5.19 shadow_geometry

```
ShadowGeometry jeod::RadiationThirdBody::shadow_geometry {Conical}
```

Flag indicating cylindrical / conical shadow geometry.

trick_units(-)

Definition at line 106 of file radiation_third_body.hh.

Referenced by calculate_shadow(), and convert_shadow_from_int().

8.12.5.20 source_to_third_hat_inrtl

```
double jeod::RadiationThirdBody::source_to_third_hat_inrtl[3] {} [protected]
```

Unit vector of source_to_third.

trick_units(-)

Definition at line 225 of file radiation_third_body.hh.

Referenced by calculate_shadow(), and update_third_body_state().

8.12.5.21 `source_to_third_inrtl`

```
double jeod::RadiationThirdBody::source_to_third_inrtl[3] {} [protected]
```

Vector from the primary source to the [RadiationThirdBody](#) in inertial RF.

trick_units(m)

Definition at line 215 of file `radiation_third_body.hh`.

Referenced by `calculate_shadow()`, and `update_third_body_state()`.

8.12.5.22 `third_to_cg_inrtl`

```
double jeod::RadiationThirdBody::third_to_cg_inrtl[3] {} [protected]
```

Vector from [RadiationThirdBody](#) to vehicle center of gravity, in inertial RF.

trick_units(m)

Definition at line 209 of file `radiation_third_body.hh`.

Referenced by `calculate_shadow()`.

8.12.5.23 `update_interval`

```
double jeod::RadiationThirdBody::update_interval {}
```

Time interval between third body state updates.

trick_units(s)

Definition at line 128 of file `radiation_third_body.hh`.

Referenced by `test_for_state_update()`.

The documentation for this class was generated from the following files:

- [radiation_third_body.hh](#)
- [radiation_third_body.cc](#)

Chapter 9

File Documentation

9.1 flat_plate_radiation_facet.cc File Reference

Define member functions for class FlatPlateRadiationFacet.

```
#include "utils/math/include/vector3.hh"
#include "utils/message/include/message_handler.hh"
#include "utils/surface_model/include/flat_plate.hh"
#include "../include/flat_plate_radiation_facet.hh"
#include "../include/radiation_messages.hh"
#include "../include/radiation_third_body.hh"
```

Namespaces

- [jeod](#)

Namespace jeod.

9.1.1 Detailed Description

Define member functions for class FlatPlateRadiationFacet.

9.2 flat_plate_radiation_facet.hh File Reference

Individual facets for use with rad environment interaction models.

```
#include "utils/sim_interface/include/jeod_class.hh"
#include "radiation_facet.hh"
```

Data Structures

- class [jeod::FlatPlateRadiationFacet](#)

A flat plate facet to be used for radiation interaction.

Namespaces

- [jeod](#)

Namespace jeod.

9.2.1 Detailed Description

Individual facets for use with rad environment interaction models.

9.3 flat_plate_radiation_factory.cc File Reference

Factory that creates a FlatPlateRadiationFacet, from a facet model.

```
#include <cstdint>
#include <typeinfo>
#include "utils/memory/include/jeod_alloc.hh"
#include "utils/message/include/message_handler.hh"
#include "utils/surface_model/include/flat_plate_thermal.hh"
#include "../include/flat_plate_radiation_facet.hh"
#include "../include/flat_plate_radiation_factory.hh"
#include "../include/radiation_messages.hh"
#include "../include/radiation_params.hh"
```

Namespaces

- [jeod](#)

Namespace jeod.

9.3.1 Detailed Description

Factory that creates a FlatPlateRadiationFacet, from a facet model.

9.4 flat_plate_radiation_factory.hh File Reference

Factory that creates an interaction facet, for a specific environment interaction model, from a facet model.

```
#include "utils/sim_interface/include/jeod_class.hh"
#include "utils/surface_model/include/interaction_facet_factory.hh"
#include "flat_plate_radiation_facet.hh"
```

Data Structures

- class [jeod::FlatPlateRadiationFactory](#)

The factory for building flat plate radiation facets.

Namespaces

- [jeod](#)

Namespace jeod.

9.4.1 Detailed Description

Factory that creates an interaction facet, for a specific environment interaction model, from a facet model.

9.5 radiation_base_facet.cc File Reference

Define member functions for class RadiationBaseFacet.

```
#include "utils/math/include/vector3.hh"
#include "utils/message/include/message_handler.hh"
#include "../include/radiation_base_facet.hh"
#include "../include/radiation_messages.hh"
#include "../include/radiation_third_body.hh"
```

Namespaces

- [jeod](#)

Namespace jeod.

9.5.1 Detailed Description

Define member functions for class RadiationBaseFacet.

9.6 radiation_base_facet.hh File Reference

Individual facets for use with radiation environment interaction models.

```
#include "interactions/thermal_rider/include/thermal_facet_rider.hh"
#include "utils/sim_interface/include/jeod_class.hh"
#include "utils/surface_model/include/interaction_facet.hh"
```

Data Structures

- class [jeod::RadiationBaseFacet](#)

Generic type of facet for radiation pressure.

Namespaces

- [jeod](#)

Namespace jeod.

9.6.1 Detailed Description

Individual facets for use with radiation environment interaction models.

9.7 radiation_default_surface.cc File Reference

Default surface for use with Radiation Pressure interaction model.

```
#include "dynamics/dyn_body/include/dyn_body.hh"
#include "interactions/thermal_rider/include/thermal_facet_rider.hh"
#include "utils/math/include/vector3.hh"
#include "utils/message/include/message_handler.hh"
#include "../include/radiation_default_surface.hh"
#include "../include/radiation_messages.hh"
#include "../include/radiation_third_body.hh"
```

Namespaces

- [jeod](#)

Namespace jeod.

9.7.1 Detailed Description

Default surface for use with Radiation Pressure interaction model.

9.8 radiation_default_surface.hh File Reference

Individual facets for use with radiation environment interaction models.

```
#include <string>
#include <utility>
#include "interactions/thermal_rider/include/thermal_facet_rider.hh"
#include "utils/sim_interface/include/jeod_class.hh"
#include "radiation_base_facet.hh"
```

Data Structures

- class [jeod::RadiationDefaultSurface](#)

The default spherical surface for radiation pressure.

Namespaces

- [jeod](#)

Namespace jeod.

9.8.1 Detailed Description

Individual facets for use with radiation environment interaction models.

9.9 radiation_facet.cc File Reference

Individual facets for use with Radiation Pressure interaction models.

```
#include "interactions/thermal_rider/include/thermal_facet_rider.hh"
#include "utils/message/include/message_handler.hh"
#include "utils/surface_model/include/facet.hh"
#include "../include/radiation_facet.hh"
#include "../include/radiation_messages.hh"
#include "../include/radiation_params.hh"
```

Namespaces

- [jeod](#)

Namespace jeod.

9.9.1 Detailed Description

Individual facets for use with Radiation Pressure interaction models.

9.10 radiation_facet.hh File Reference

Individual facets for use with radiation environment interaction models.

```
#include "interactions/thermal_rider/include/thermal_facet_rider.hh"
#include "utils/sim_interface/include/jeod_class.hh"
#include "utils/surface_model/include/interaction_facet.hh"
#include "radiation_base_facet.hh"
#include "utils/surface_model/include/facet.hh"
```

Data Structures

- class [jeod::RadiationFacet](#)

Generic type of facet for radiation pressure.

Namespaces

- [jeod](#)

Namespace jeod.

9.10.1 Detailed Description

Individual facets for use with radiation environment interaction models.

9.11 radiation_messages.cc File Reference

Implement the class RadiationMessages.

```
#include "utils/message/include/make_message_code.hh"
#include "../include/radiation_messages.hh"
```

Namespaces

- [jeod](#)

Namespace jeod.

Macros

- `#define MAKE\_RADIATION\_MESSAGE\_CODE(id) JEOD_MAKE_MESSAGE_CODE(RadiationMessages, "interactions/radiation_pressure/", id)`

9.11.1 Detailed Description

Implement the class RadiationMessages.

9.11.2 Macro Definition Documentation

9.11.2.1 MAKE_RADIATION_MESSAGE_CODE

```
#define MAKE_RADIATION_MESSAGE_CODE(  
    id ) JEOD_MAKE_MESSAGE_CODE(RadiationMessages, "interactions/radiation_pressure/",  
    id)
```

Definition at line 43 of file radiation_messages.cc.

9.12 radiation_messages.hh File Reference

Define the class RadiationMessages, the class that specifies the message IDs used in the Radiation model.

```
#include "utils/sim_interface/include/jeod_class.hh"
```

Data Structures

- class [jeod::RadiationMessages](#)
Provides error messages.

Namespaces

- [jeod](#)
Namespace jeod.

9.12.1 Detailed Description

Define the class RadiationMessages, the class that specifies the message IDs used in the Radiation model.

9.13 radiation_params.hh File Reference

A virtual base class for radiation facet parameters, used to create interaction facets for radiation in the Interaction↔SurfaceFactorys.

```
#include "interactions/thermal_rider/include/thermal_params.hh"  
#include "utils/sim_interface/include/jeod_class.hh"  
#include "utils/surface_model/include/facet_params.hh"
```

Data Structures

- class [jeod::RadiationParams](#)
Provides a parameter list to each facet, based on the facet material properties.

Namespaces

- [jeod](#)
Namespace jeod.

9.13.1 Detailed Description

A virtual base class for radiation facet parameters, used to create interaction facets for radiation in the Interaction↔SurfaceFactorys.

9.14 radiation_pressure.cc File Reference

Calculation of force and torque due to radiation pressure.

```
#include <cstdlib>
#include "dynamics/dyn_manager/include/dyn_manager.hh"
#include "interactions/thermal_rider/include/thermal_facet_rider.hh"
#include "utils/math/include/vector3.hh"
#include "utils/memory/include/jeod_alloc.hh"
#include "utils/message/include/message_handler.hh"
#include "../include/radiation_messages.hh"
#include "../include/radiation_pressure.hh"
#include "../include/radiation_source.hh"
#include "../include/radiation_third_body.hh"
```

Namespaces

- [jeod](#)

Namespace jeod.

9.14.1 Detailed Description

Calculation of force and torque due to radiation pressure.

9.15 radiation_pressure.hh File Reference

Radiation pressure top-level definition.

```
#include "interactions/thermal_rider/include/thermal_model_rider.hh"
#include "utils/container/include/pointer_vector.hh"
#include "utils/sim_interface/include/jeod_class.hh"
#include "radiation_source.hh"
#include "radiation_default_surface.hh"
#include "radiation_surface.hh"
```

Data Structures

- class [jeod::RadiationPressure](#)

Radiation pressure top-level definition.

Namespaces

- [jeod](#)

Namespace jeod.

9.15.1 Detailed Description

Radiation pressure top-level definition.

9.16 radiation_pressure__default_surface.cc File Reference

Calculation of force and torque due to radiation pressure.

```
#include "dynamics/dyn_manager/include/dyn_manager.hh"
#include "interactions/thermal_rider/include/thermal_model_rider.hh"
#include "utils/math/include/vector3.hh"
#include "../include/radiation_default_surface.hh"
#include "../include/radiation_pressure.hh"
#include "../include/radiation_source.hh"
#include "../include/radiation_third_body.hh"
```

Namespaces

- [jeod](#)

Namespace jeod.

9.16.1 Detailed Description

Calculation of force and torque due to radiation pressure.

9.17 radiation_pressure__surface_model.cc File Reference

Calculation of force and torque due to radiation pressure.

```
#include "dynamics/dyn_manager/include/dyn_manager.hh"
#include "interactions/thermal_rider/include/thermal_model_rider.hh"
#include "utils/math/include/vector3.hh"
#include "../include/radiation_pressure.hh"
#include "../include/radiation_source.hh"
#include "../include/radiation_surface.hh"
#include "../include/radiation_third_body.hh"
```

Namespaces

- [jeod](#)

Namespace jeod.

9.17.1 Detailed Description

Calculation of force and torque due to radiation pressure.

9.18 radiation_source.cc File Reference

Definition of methods associated with Primary Sources.

```
#include <cstdint>
#include "dynamics/dyn_manager/include/dyn_manager.hh"
#include "environment/planet/include/planet.hh"
#include "utils/math/include/vector3.hh"
#include "utils/memory/include/jeod_alloc.hh"
#include "utils/message/include/message_handler.hh"
#include "utils/ref_frames/include/ref_frame.hh"
#include "../include/radiation_source.hh"
#include "../include/radiation_messages.hh"
```

Namespaces

- [jeod](#)

Namespace jeod.

9.18.1 Detailed Description

Definition of methods associated with Primary Sources.

9.19 radiation_source.hh File Reference

Radiation pressure parameter and variable definitions.

```
#include <string>
#include <utility>
#include "utils/sim_interface/include/jeod_class.hh"
```

Data Structures

- class [jeod::RadiationSource](#)

Provides information on the source of the incident radiation.

Namespaces

- [jeod](#)

Namespace jeod.

9.19.1 Detailed Description

Radiation pressure parameter and variable definitions.

9.20 radiation_surface.cc File Reference

Vehicle surface model for Radiation Pressure model.

```
#include <cstdint>
#include "dynamics/dyn_body/include/dyn_body.hh"
#include "utils/math/include/vector3.hh"
#include "utils/memory/include/jeod_alloc.hh"
#include "utils/message/include/message_handler.hh"
#include "utils/surface_model/include/facet.hh"
#include "utils/surface_model/include/interaction_facet.hh"
#include "utils/surface_model/include/interaction_facet_factory.hh"
#include "../include/radiation_facet.hh"
#include "../include/radiation_messages.hh"
#include "../include/radiation_surface.hh"
```

Namespaces

- [jeod](#)

Namespace jeod.

9.20.1 Detailed Description

Vehicle surface model for Radiation Pressure model.

9.21 radiation_surface.hh File Reference

Vehicle surface model for general environment interaction models.

```
#include "utils/sim_interface/include/jeod_class.hh"
#include "utils/surface_model/include/interaction_surface.hh"
#include "radiation_facet.hh"
```

Data Structures

- class [jeod::RadiationSurface](#)

The surface of the vehicle that interacts with the incident flux.

Namespaces

- [jeod](#)

Namespace jeod.

9.21.1 Detailed Description

Vehicle surface model for general environment interaction models.

9.22 radiation_surface_factory.cc File Reference

Factory that creates an interaction surface, for a specific environment interaction model, from a surface model.

```
#include <cstdint>
#include "utils/message/include/message_handler.hh"
#include "../include/radiation_messages.hh"
#include "../include/radiation_params.hh"
#include "../include/radiation_surface_factory.hh"
```

Namespaces

- [jeod](#)

Namespace jeod.

9.22.1 Detailed Description

Factory that creates an interaction surface, for a specific environment interaction model, from a surface model.

9.23 radiation_surface_factory.hh File Reference

Factory that creates an interaction surface, for a specific environment interaction model, from a surface model.

```
#include "utils/sim_interface/include/jeod_class.hh"
#include "utils/surface_model/include/interaction_surface_factory.hh"
#include "flat_plate_radiation_factory.hh"
```

Data Structures

- class [jeod::RadiationSurfaceFactory](#)

The factory for creating Radiation Surfaces.

Namespaces

- [jeod](#)

Namespace jeod.

9.23.1 Detailed Description

Factory that creates an interaction surface, for a specific environment interaction model, from a surface model.

9.24 radiation_third_body.cc File Reference

Provides the functionality associated with RadiationThirdBodys.

```
#include <cstdint>
#include "dynamics/dyn_body/include/dyn_body.hh"
#include "dynamics/dyn_manager/include/dyn_manager.hh"
#include "environment/planet/include/planet.hh"
#include "utils/math/include/vector3.hh"
#include "utils/message/include/message_handler.hh"
#include "../include/radiation_messages.hh"
#include "../include/radiation_source.hh"
#include "../include/radiation_third_body.hh"
```

Namespaces

- [jeod](#)

Namespace jeod.

9.24.1 Detailed Description

Provides the functionality associated with RadiationThirdBodys.

9.25 radiation_third_body.hh File Reference

Define the class RadiationThirdBody.

```
#include <string>
#include <utility>
#include "utils/sim_interface/include/jeod_class.hh"
```

Data Structures

- class [jeod::RadiationThirdBody](#)

Provide information on bodies that may cause shadowing or reflected illumination.

Namespaces

- [jeod](#)

Namespace jeod.

9.25.1 Detailed Description

Define the class RadiationThirdBody.

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